GE Energy

GEK-SA-1031A

Mark* VIe Control

Initial Startup for Energy Products Europe (EPE)

Instruction Guide

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Introduction

The Mark* VIe controller runs the application code. The controller operating system (OS) is QNX® Neutrino™ a real time, multitasking OS designed for high-speed, high reliability industrial applications. Communication ports provide links to I/O and engineering interfaces as follows:

- Ethernet connection ENET1 for the Unit Data Highway (UDH) communication with HMIs, and other control equipment
- Ethernet connection for the R, S, and T I/O network
- RS-232C connection for setup using COM1 port
- RS-232C connection for Modbus® communication using COM2 port
- Ethernet connection ENET2 (optional) for Modbus communication with thirdparty systems

This document provides instructions for the first-time power on, initial download, and troubleshooting of the Mark VIe control in a TMR configuration for gas turbine, steam turbine, and BOP (DCS) applications. Some or all boards discussed could be present in the Mark VIe system. Depending on the system redundancy, downloading should be performed for the controllers and the I/O packs as appropriate. Troubleshooting steps could be different for systems in dual or simplex configurations.

Note This document is intended for use by GE Energy System Integrators and Field engineers for startup and troubleshooting issues related to the Mark VIe control system. The document is recommended for use by qualified personnel with functional knowledge of the ToolboxST* application and the Mark VIe control system.

Mark VIe Initial Power-on Procedure

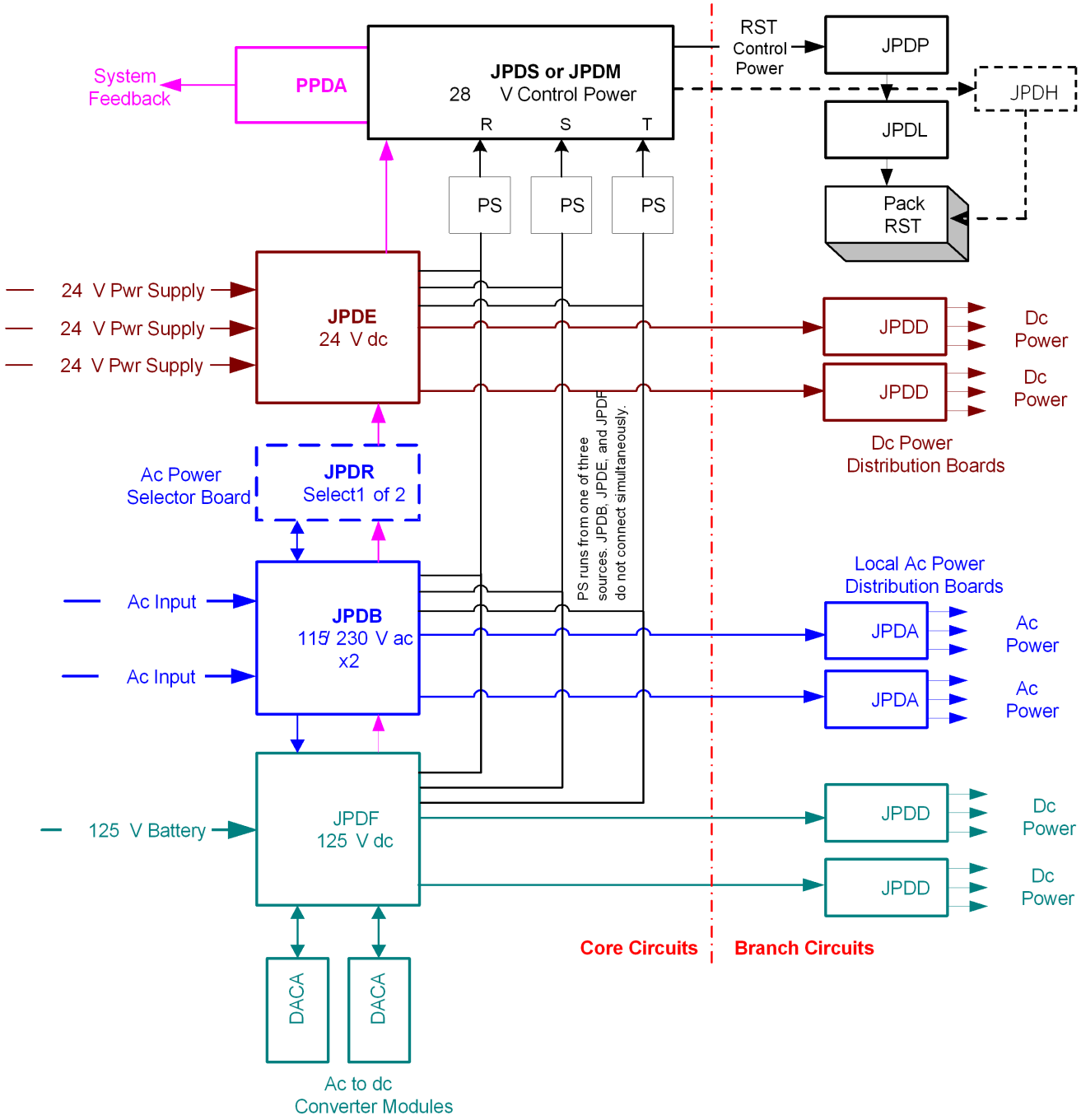
The Power Distribution Modules (PDM) are designed specifically for Mark VIe control. The PDM uses individual boards to accept and condition primary control power inputs of 125 V dc, 24 V dc, and 115/230 V ac for use in redundant combinations. Applied power is distributed to system terminal boards for use in field circuits, and converted to 28 V dc to operate Mark VIe I/O packs, Ethernet switches and controllers (UCSA only).

The PDM is divided into two different categories:

Core Circuit distribution circuits serve as the primary power management for a cabinet or series of cabinets. Input power from one or more sources is received by a corresponding module or board. The power is distributed to terminal boards, as well as one or more bulk power supplies producing 28 V dc power to operate the control electronics. The 28 V power is monitored and distributed by one or more 28 V output boards. Ac input power is received by a JPDB module, which accepts two independent ac sources. Dc input power is managed by the JPDE (24 V/48 V) and JPDF (125 V) modules.

The 28 V dc control power output board (JPDS or JPDM) hosts a PPDA I/O pack providing system feedback. Ribbon cables can daisy-chain other core boards in the system to the board holding the PPDA I/O pack. The PPDA produces system feedback signals for all power bus voltages, branch circuit status, ground fault detection, and bulk power supply health. Complete monitoring and system feedback sets this power system apart from conventional methods of power distribution.

Branch Circuit boards split the power output from the PDM core components into individual ac and dc circuits for use in the cabinets. Branch circuits do not connect to the PPDA I/O pack for system feedback. Elements receiving power from the branch circuits provide their own power status feedback signals to the control system. Branch circuit elements are usually single circuit boards rather than modules.



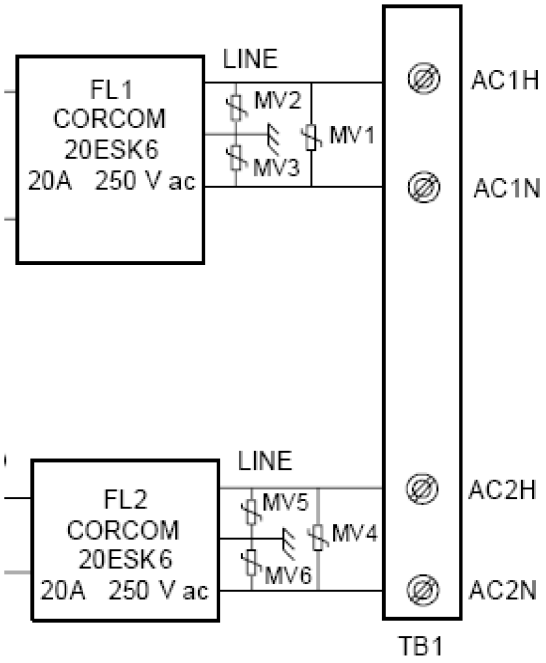
Power Distribution Module (PDM) Basic Layout

The Mark VIe control initial power-on should be completed inside the TCC (Turbine Control Cubicle) with the air conditioner turned on in a dust-free environment (controllers, I/O packs and IONet switches are sensitive to dust).

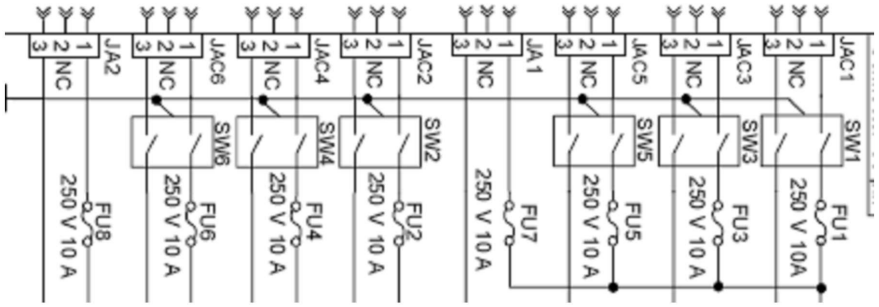
➢ To apply power to the Mark VIe cabinet

1. Verify that the JPDB terminal board is properly configured.

a. Verify that the ac1 supply is connected to the AC1H and AC1N terminals, and that the ac2 supply source (optional) is connected to the AC2H and AC2N terminals.



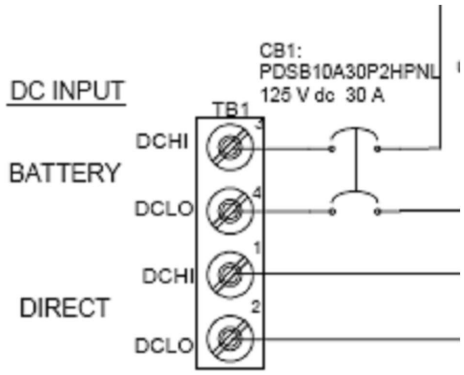
b. Turn off switches SW1 through SW6.



c. Remove all the JACx cables from their connectors.

2. Verify that the JPDF is properly configured.

   a. Verify that the 125 V dc battery connections are done at terminals 1 and 2 of the TB1, and dc direct connections are done at terminals 3 and 4 of the TB1.



   b. Verify that the SW1R, SW1S, SW1T, SW1X, SW1Y and SW1Z switches are turned off.

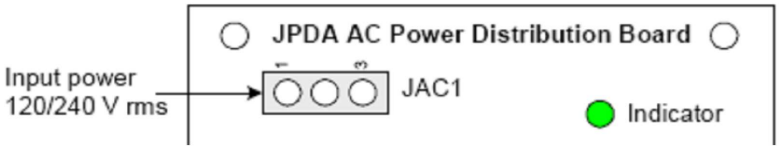
   c. Remove the J12, J8A, J8B, JR1, JS1, JT1, J7X, J7Y, and J7Z cables from the connector.

3. Apply ac power supply to the JPDB.

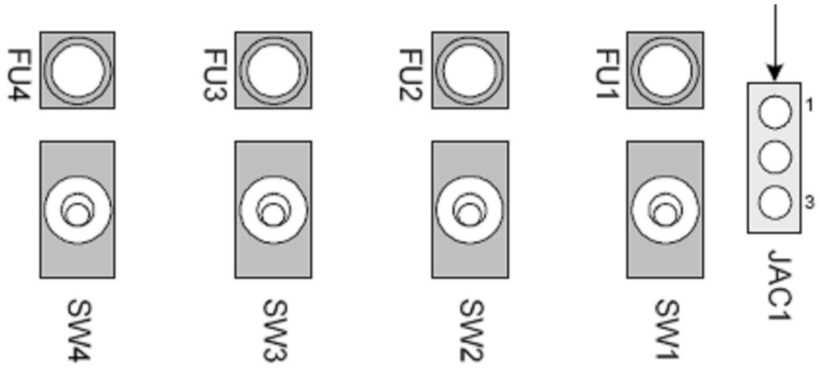
- Measure the ac voltage at the JPDB terminals AC1H/ AC1N and AC2H/ AC2N to verify that they are within desired limits.

4. Verify that the JPDA is properly configured.

   a. Verify that the JACx cable is connected to the JACx connector on the JPDA.

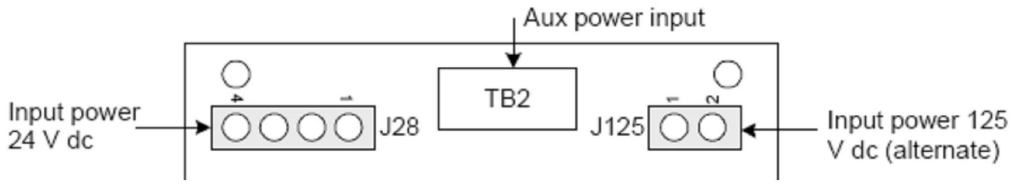


   b. Verify that switches SW1 through SW4 are off.

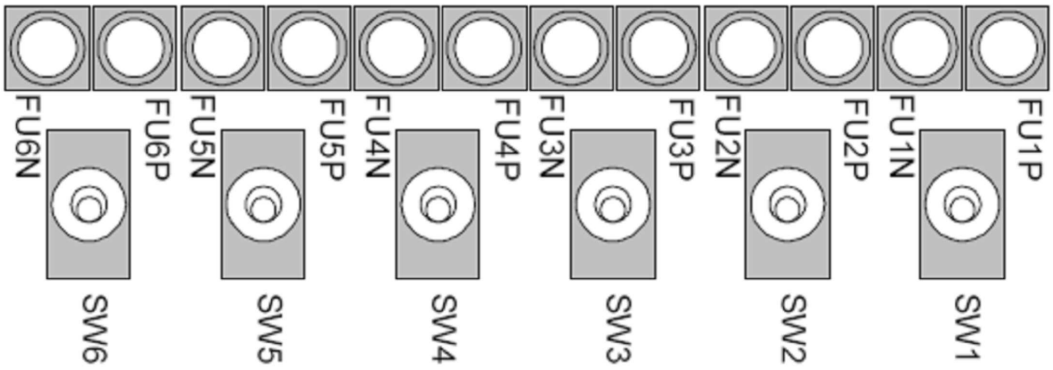


5. Verify that the JPDD is properly configured.

   a. Verify that the J28 cable is attached to the connector.



   b. Verify that the SW1 through SW6 switches are off.



6. Verify that the vendor-manufactured control power supplies used by the Mark VIe control are properly configured.

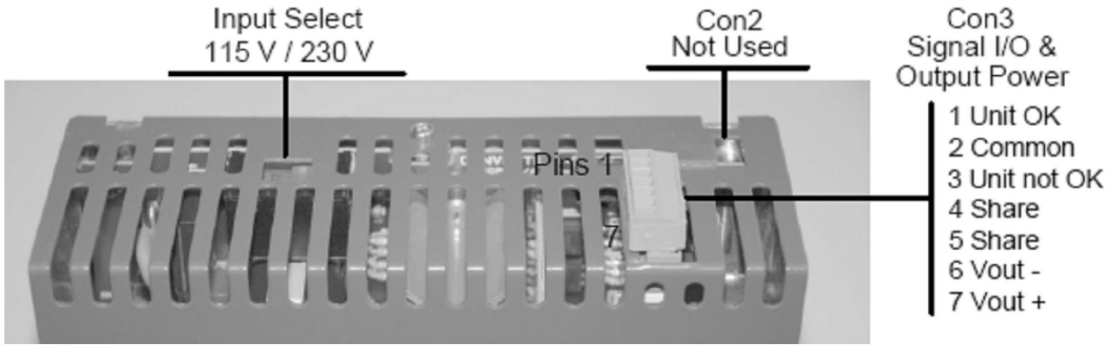
The Mark VIe control uses several vendor-manufactured control power supplies, which provide the following features:

- Convection cooling – no cooling fans used
- Ambient temperature range of -30ºC to 65ºC (-22 ºF to 149 ºF)
- A ± 2% voltage regulation for 24, 28, and 48 V dc output
- Compatibility with Mark VIe vibration and contamination requirements
- An open dry contact for status feedback
- Support for parallel operation without extra components. Diode equivalent is included on each power supply outlet
- Load share of multiple supplies when wired together
- Current limit and over-voltage protection of outputs
- Input filtering to prevent sensitivity to input interference
- CE-marked supplies

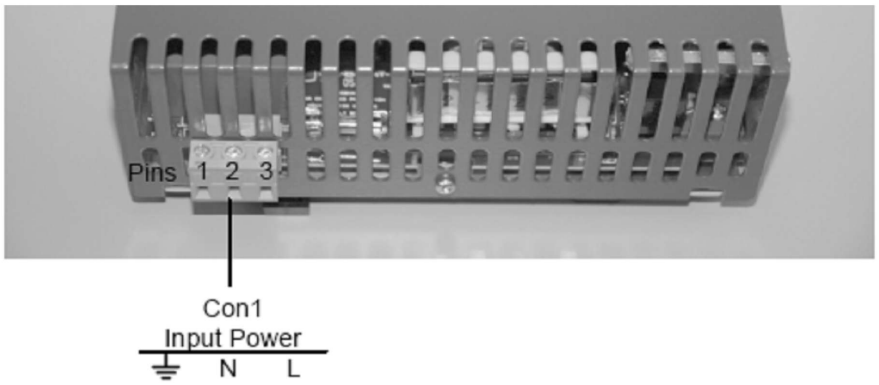
a. Physically verify the power supply installed in the cabinet, and confirm that the connections are configured correctly.

b. Verify the connection for the following power supply:

- **Ac Input 150 W 28 V dc Power Supply – GE Part No. 342A4917P150W28**



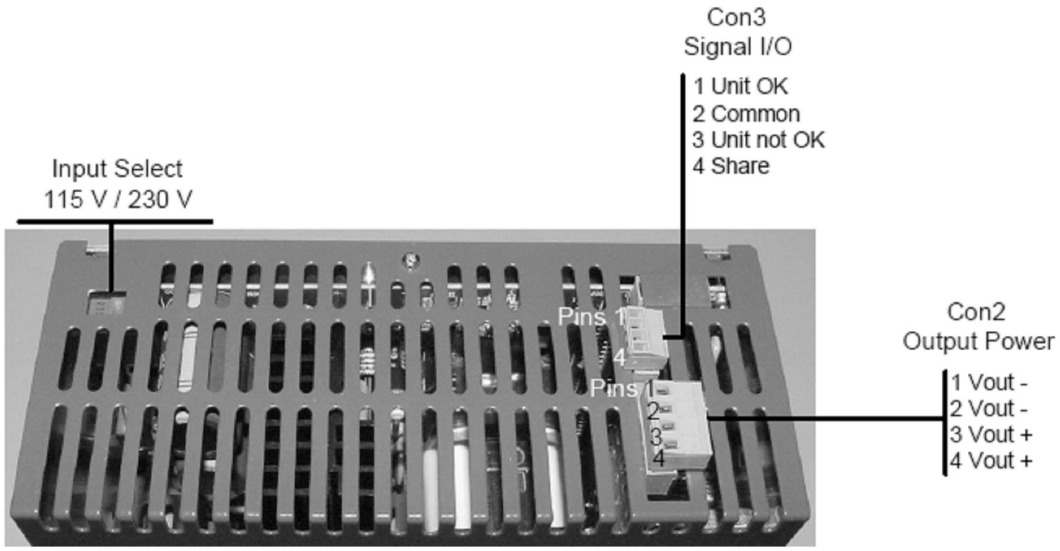
Connection on Top of Power Supply



Connection on Bottom of Power Supply

c. Verify the connection for the following power supplies:

- **Ac Input 300 W 24 V dc Power Supply – 342A4917P300W24**
- **Ac Input 300 W 28 V dc Power Supply – 336A4940FEP01**



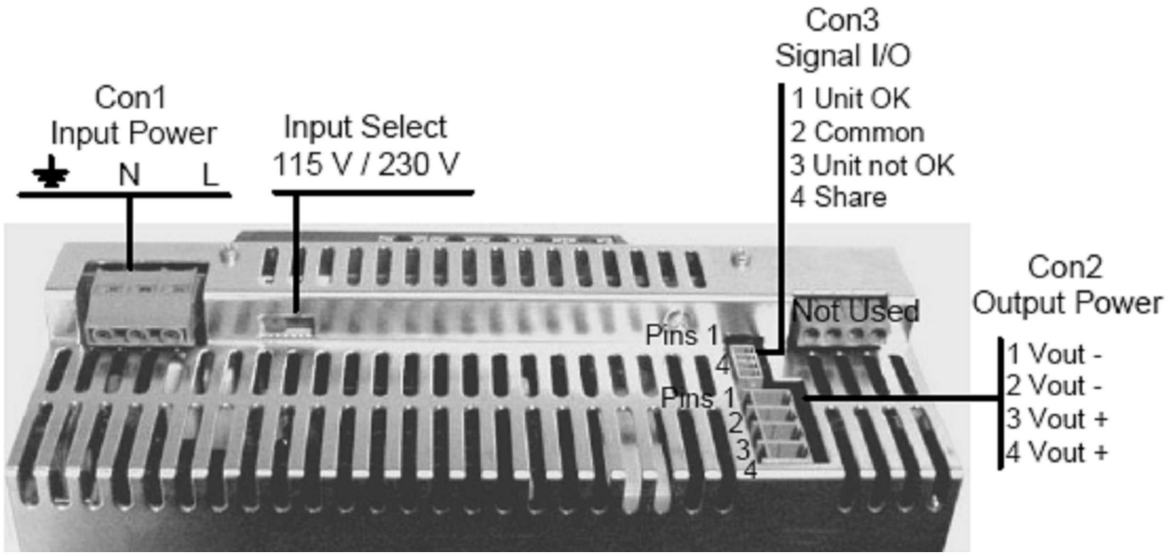
Connection on Top of Power Supply



Connection on Bottom of Power Supply

Verify the connection for the following power supplies:

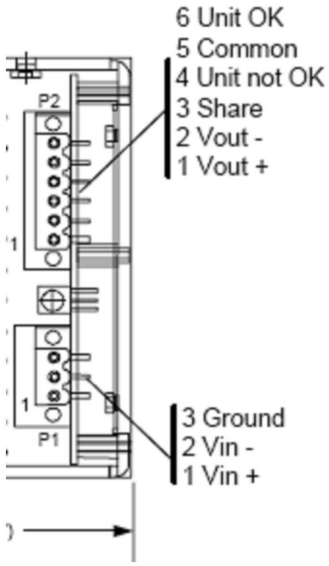
- **Ac Input 600 W 24 V dc Power Supply – 342A4917P600W24**
- **Ac Input 600 W 28 V dc Power Supply – 342A4917P600W28**
- **Ac Input 600 W 48 V dc Power Supply – 342A4917P600W48**



Connection on Top of Power Supply

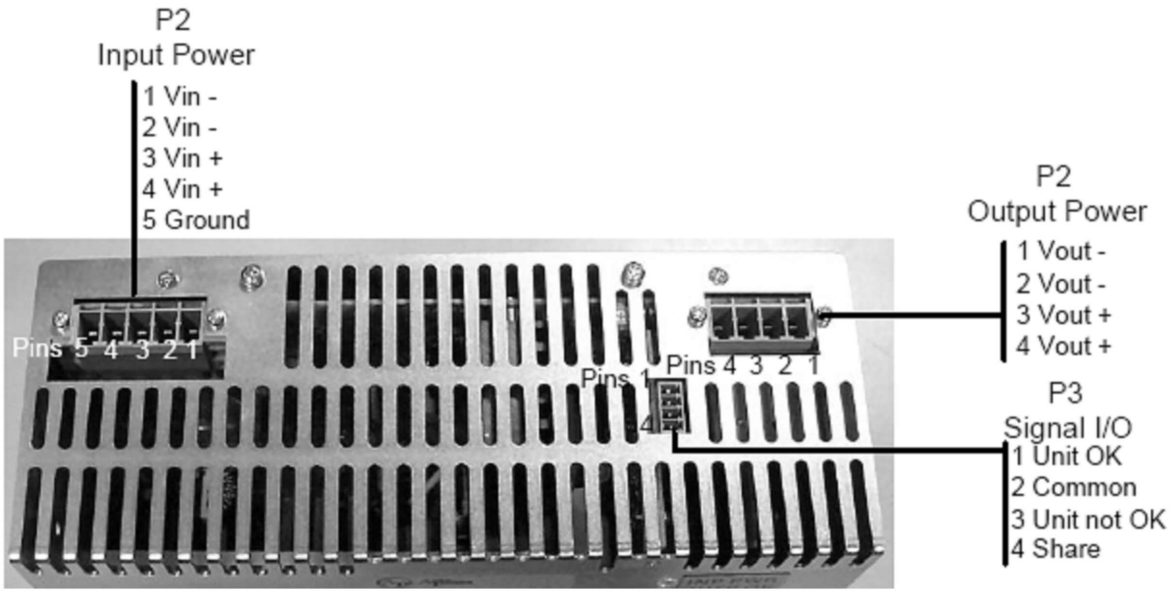
d. Verify the connection for the following power supplies:

- **24 V dc In/150 W 28 V dc Power Supply – 342A4922P28V150DLA**
- **125 V dc In/150 W 28 V dc Power Supply – 342A4922P28V150DHA**



e. Verify the connection for the following power supplies:

- **24 V dc In/500 W 28 V dc Power Supply – 342A4922P28V500DL**
- **125 V dc In/500 W 28 V dc Power Supply – 342A4922P28V500DH**



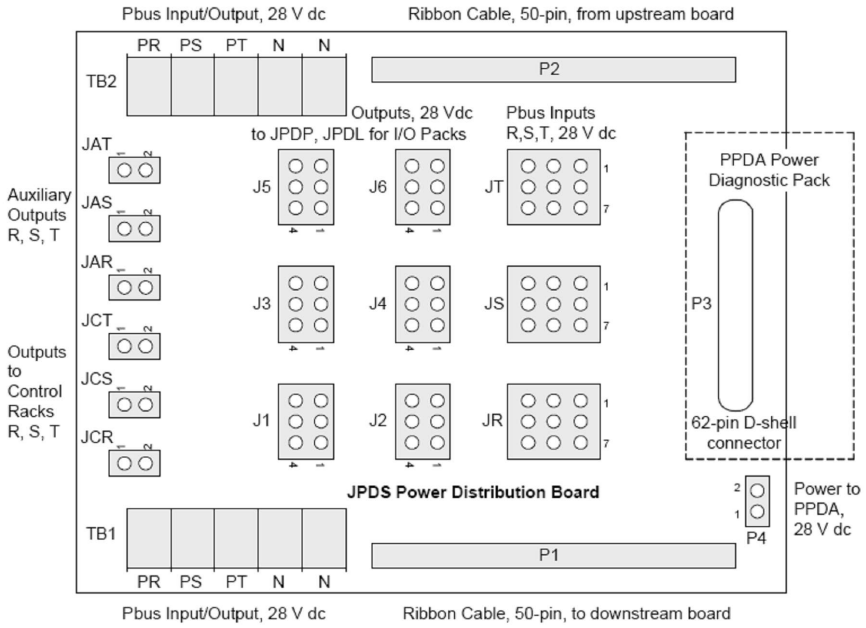
7. Verify that the JPDS and the JPDM are configured correctly.

a. Verify that the cables (JR JS and JT) from the vendor-manufactured control power supplies are correctly connected.

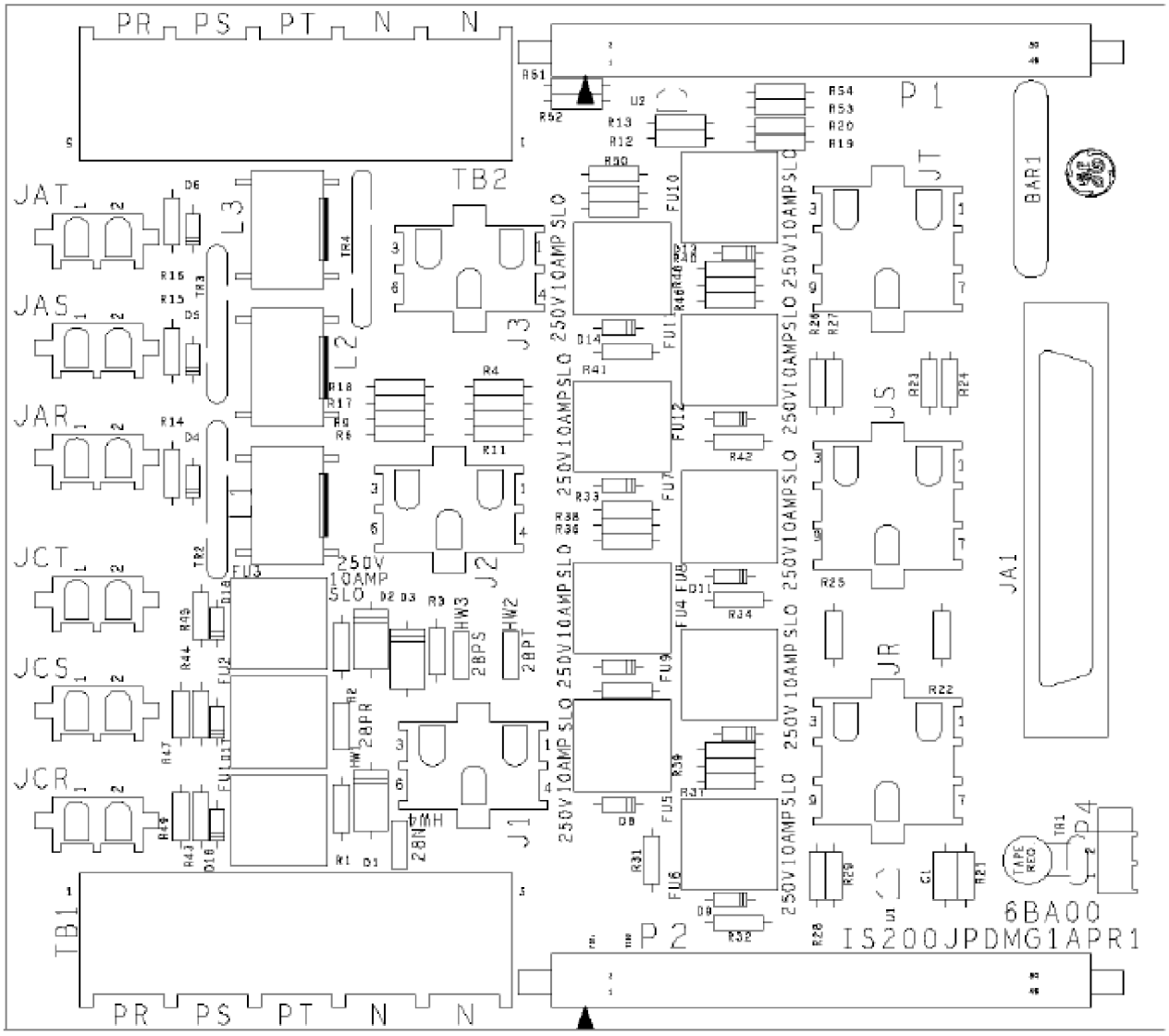
b. Verify that cables J1- J6 are correctly connected.

c. Verify that cables JAR, JAS, JAT, JCR, JCS and JCT are correctly connected.

d. The PPDA is positioned at the P3 location, and receives power from the P4 connector.

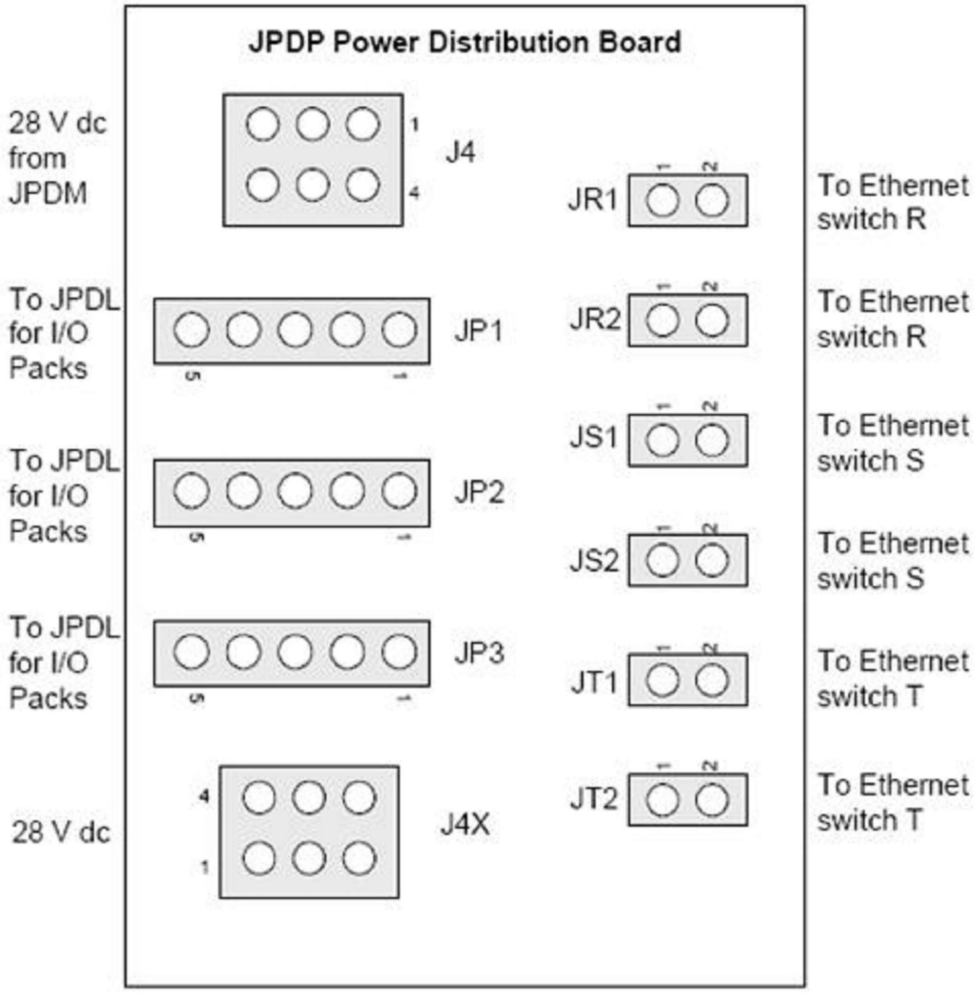


JPDS Layout



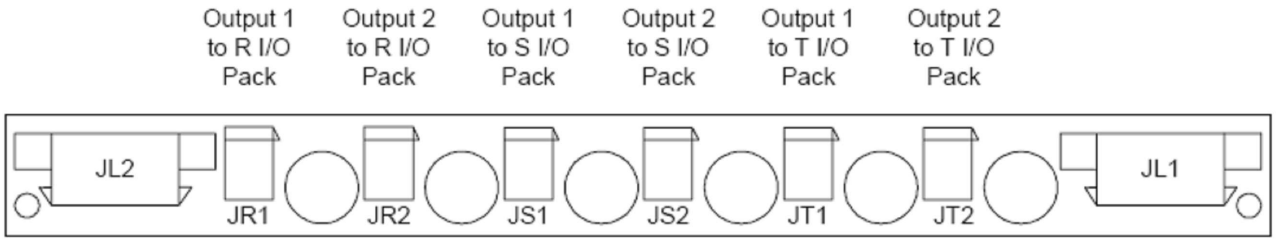
JPDM Mechanical Board Layout

8. Verify that the JPDP is properly configured and all connections complete. Then remove the cables from the JR1, JR2, JS1, JS2, JT1, and JT2 connectors.



Note Follow the above step for all JPDP boards.

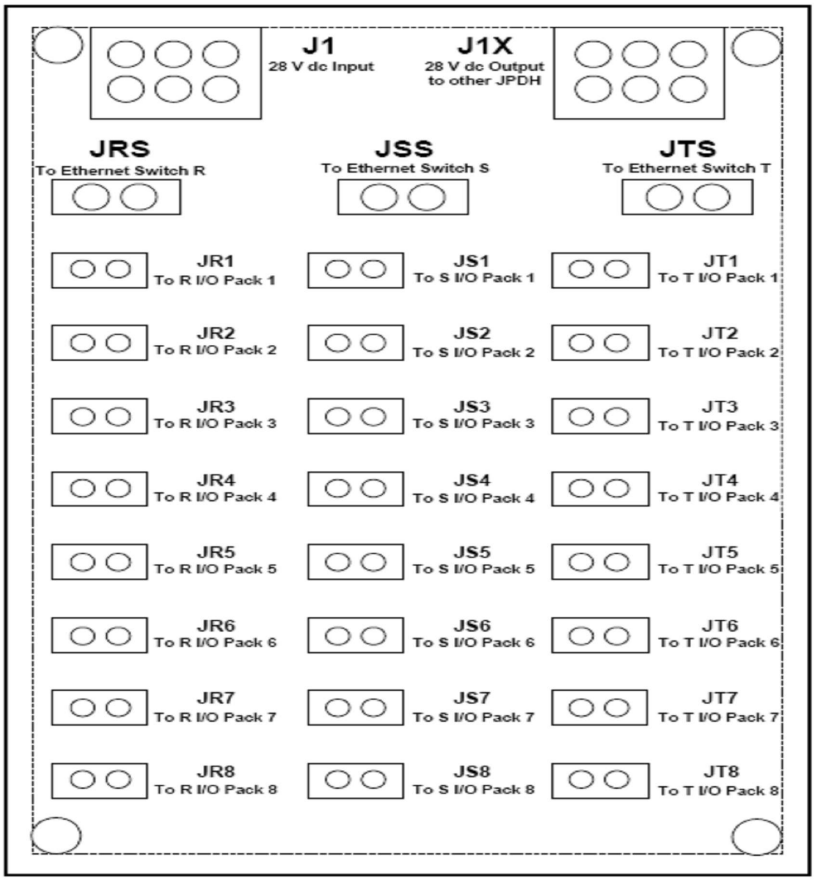
9. Verify that the the JPDL is correctly configured, verify that all connections on the JPCL are complete, then remove the JR1, JR2, JS1, JS2, JT1 and JT2 cables.



JPDL Cabling

Note Follow the above step for all JPDL boards.

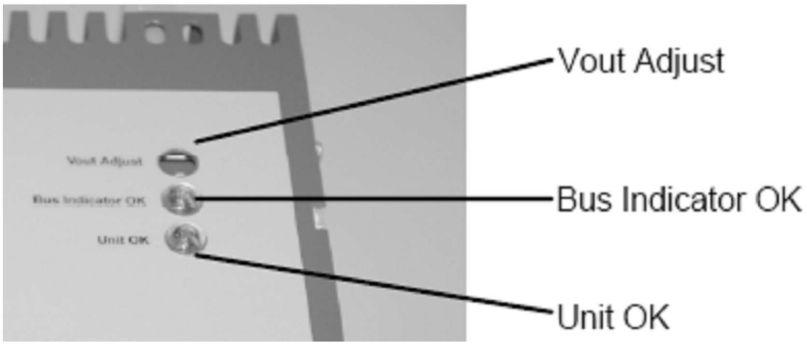
10. Verify that the JPDH is correctly configured, then verify that all connections are complete. Remove the JR1 to JR8, JS1 to JS8, and the JT1 to JT8 cables.



Note Follow the above step for all JPDH boards.

11. Apply dc power to the JPDF, and measure the dc voltage at TB1 terminals 1/2 and terminals 3/4 to ensure that voltages are within specified limits.
12. Verify the dc voltage split at the JPDF. The dc voltage should split between – 62.5 V and + 62.5 V.
13. Verify that the output voltage of J12, J8A, J8B, J1R, J1S, J1T, J7X, J7Y, and J7Z are within limits.
14. Remove the JR, JS and JT cables from the JPDS/ JPDM power supply.
15. Turn off the dc supply to JPDF, and reconnect the J12, J8A, J8B, J1R, J1S, J1T, J7X, J7Y, and J7Z cables to each respective location on JPDF.
16. Apply dc input power to the JPDF, and turn on the SW1R, SW1S, SW1T, SW7X, SW7Y and SW7Z switches on the JPDF board.

17. Verify that the vendor-provided power supply is healthy.



Note The Vout Adjust potentiometer (optional) allows adjustment of the output voltage from 24 V dc to 32 V dc. The Bus Indicator displays solid green when input power is applied. The Unit OK displays solid green when the supply is within regulation and displays solid red when the supply is unhealthy.

18. From the JPDD board, remove the JD1 to JD6 connectors, turn on the SW1- SW6 switches, and measure the power supply at each JD connector. Reconnect all JD cables to their original location after verifying that the power supply is within limits.

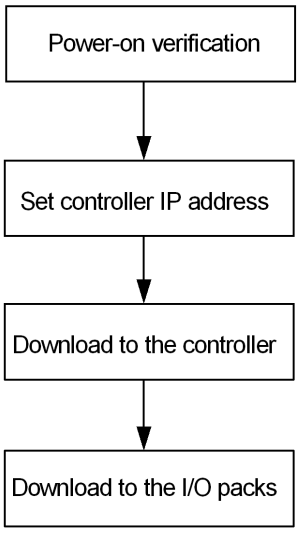
19. Turn off the SW1R, SW1S and SW1T switches, and connect the JR, JS, and JT cables from the power supply to the JPDS and the JPDM.

20. Turn on the SW1R, SW1S and SW1T switches, then verify that the JPDL is supplying the proper voltage. If voltage is correct, power down the SW1R, SW1S, and SW1T switches, then connect the JR1, JS1, JT1, JR2, JS2, and JT2 cables to verify that the I/O packs connected to the board are powered-on properly.

Downloading

A successful Mark VIe download is performed in a dust-free environment, located within the Turbine Control Cubicle (TCC) with the air conditioner running.

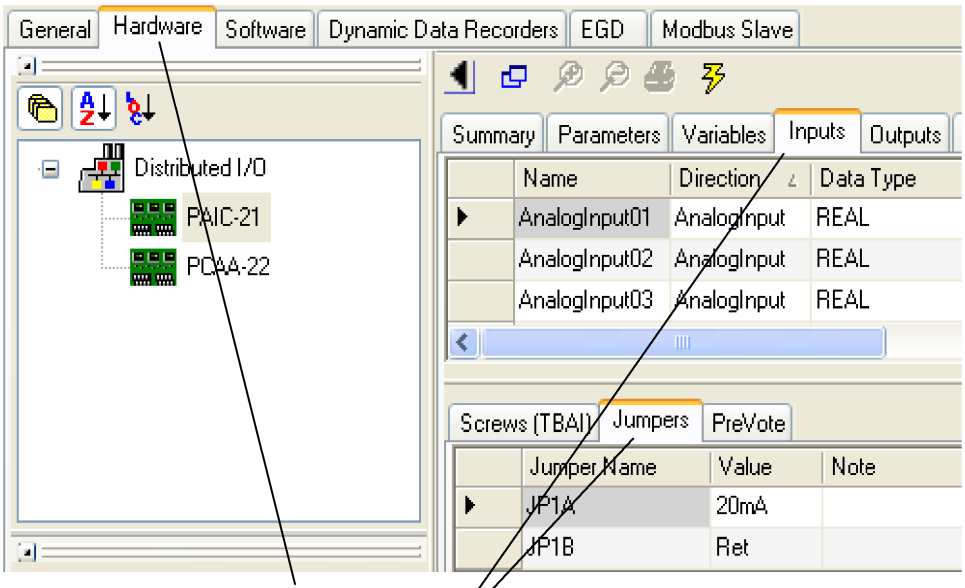
Download to the Mark VIe cabinet as follows:



Before downloading the controller, verify the following:

- IR links on packs are masked (disabled)
- Jumpers on the terminal boards are correctly configured.

Note Typically, I/O packs that have an Analog Inputs tab have jumpers.



From the **Component Editor Hardware** tab, select the desired I/O pack, then select the **Inputs** (Or **Analog Inputs**) tab to display the **Jumpers** tab.

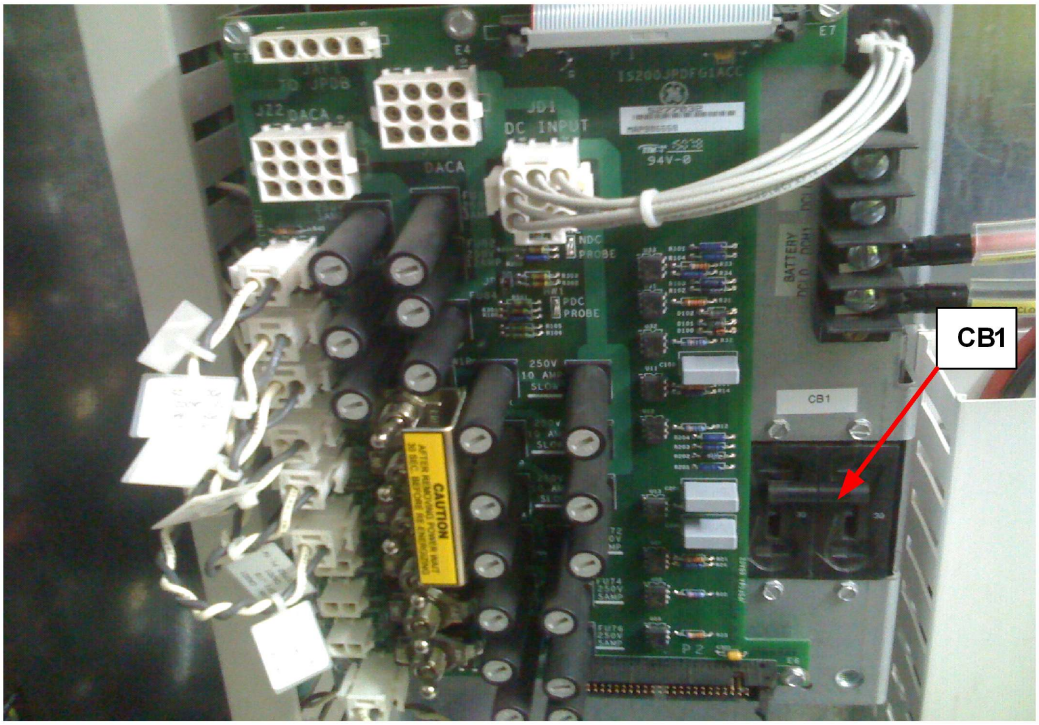
- Mark VIe project software (.tcw file) and the Modify Data password are available.

Power-on Verification (on site/integrators)

Refer to the section, Mark VIe Initial Power-on Procedure.

➢ **To apply power to the cabinet**

1. Turn on the CB1 breaker adjacent to the JPDF board.

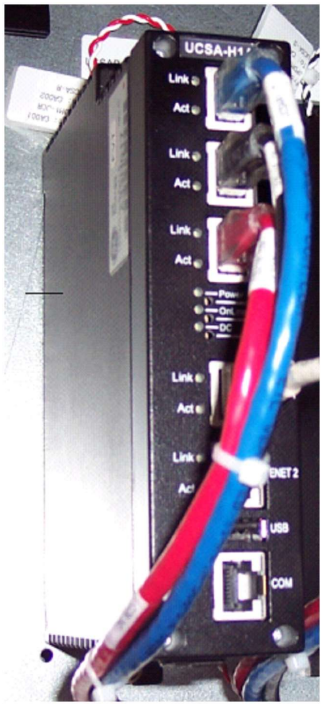


JPDF

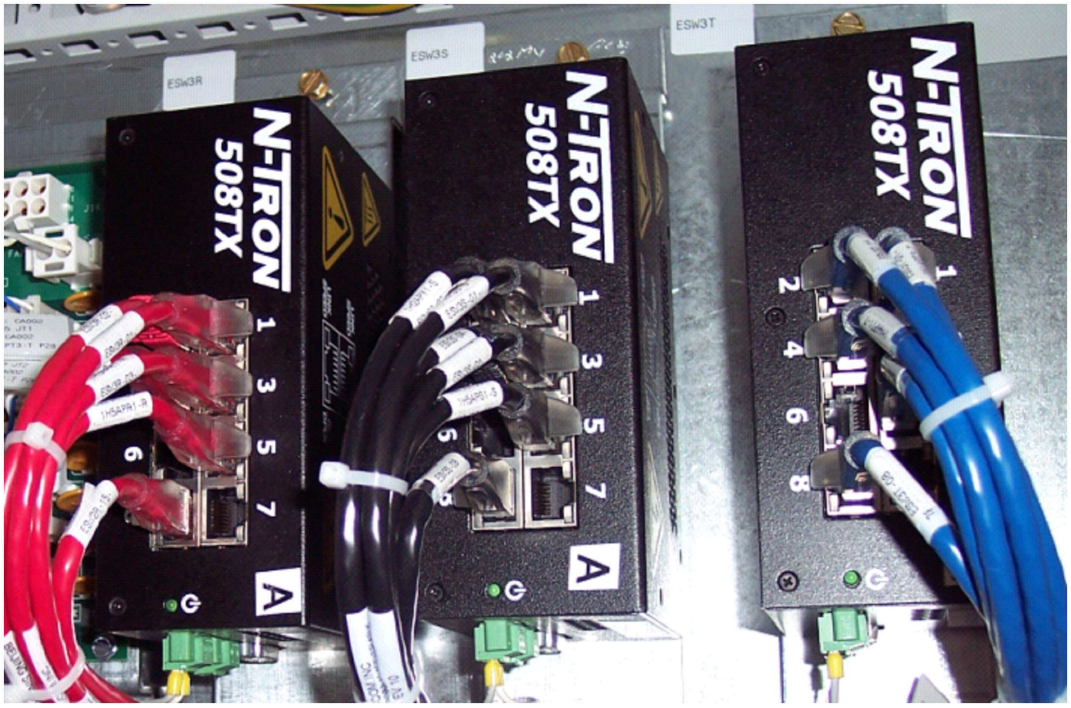
2. Verify that every component (including controller, IONet switches and I/O packs) has power applied.



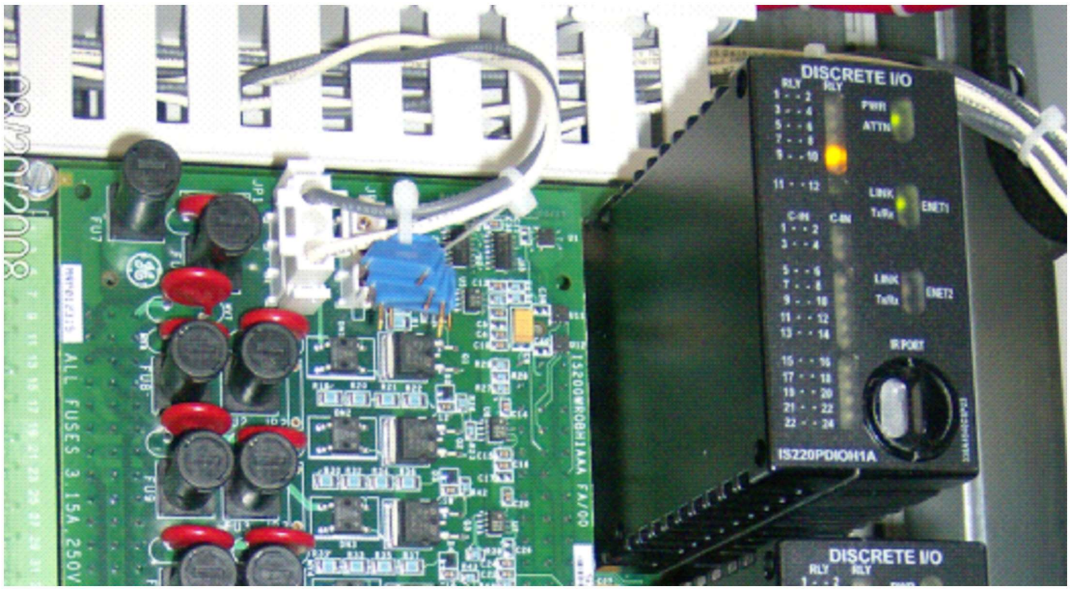
UCCA Controller



UCSA Controller



IONET Switches



I/O Packs

Setting Controller IP Address

Using Serial Cable

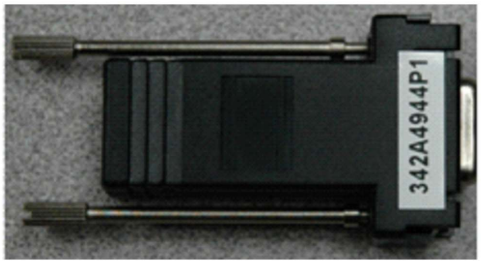
Refer to GEI-100665, Mark VIe Controllers (UCCx and UCSx) Instruction Guide, the section Specifications of each controller type for the COM port connector hardware required for additional information.

Procedures for assigning the IP address of various Mark VIe controllers are provided. When using the *Configure Network Address* option, the procedures for all controllers is the same. The UCSB controller provides an *Initialize USB* option, while all others provide a *Format Flash* option. To download a Mark VIe controller IP address and perform an initial controller download, you need the following:

For UCSA and UCSB controllers:



Standard 4 pair UTP cable (Ethernet Cable)



Controller Connector (Part #: 342A4944P1)

For UCCC controllers:



Standard 4 pair UTP cable (Ethernet Cable)



Controller Connector (Part #: 342A4931ABP1)

For UCCA and UCVx controllers:



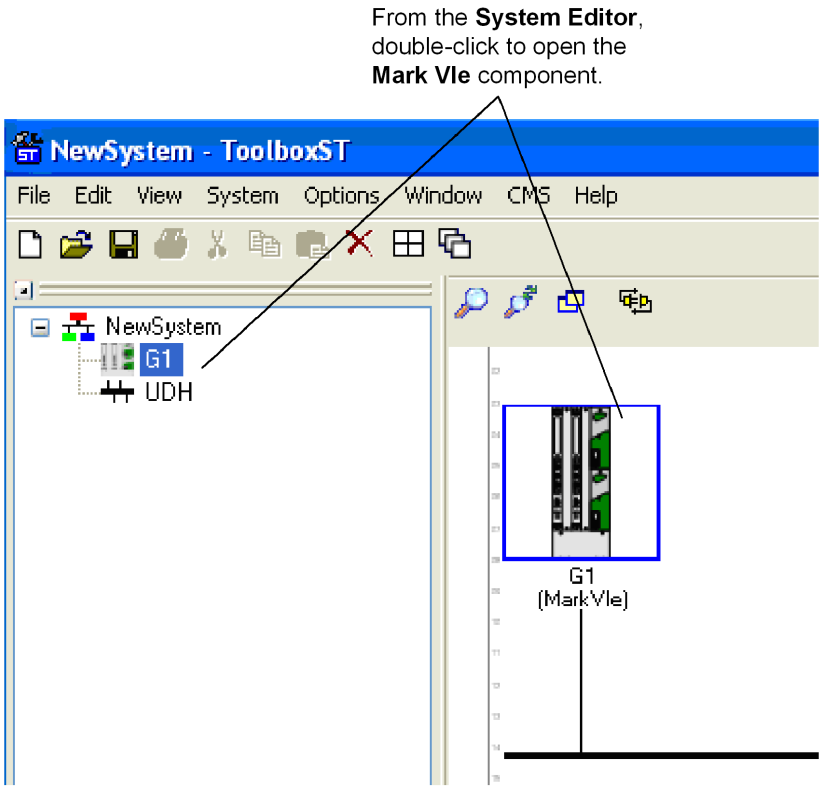
Serial Cable (Part #: 336A3582P1)



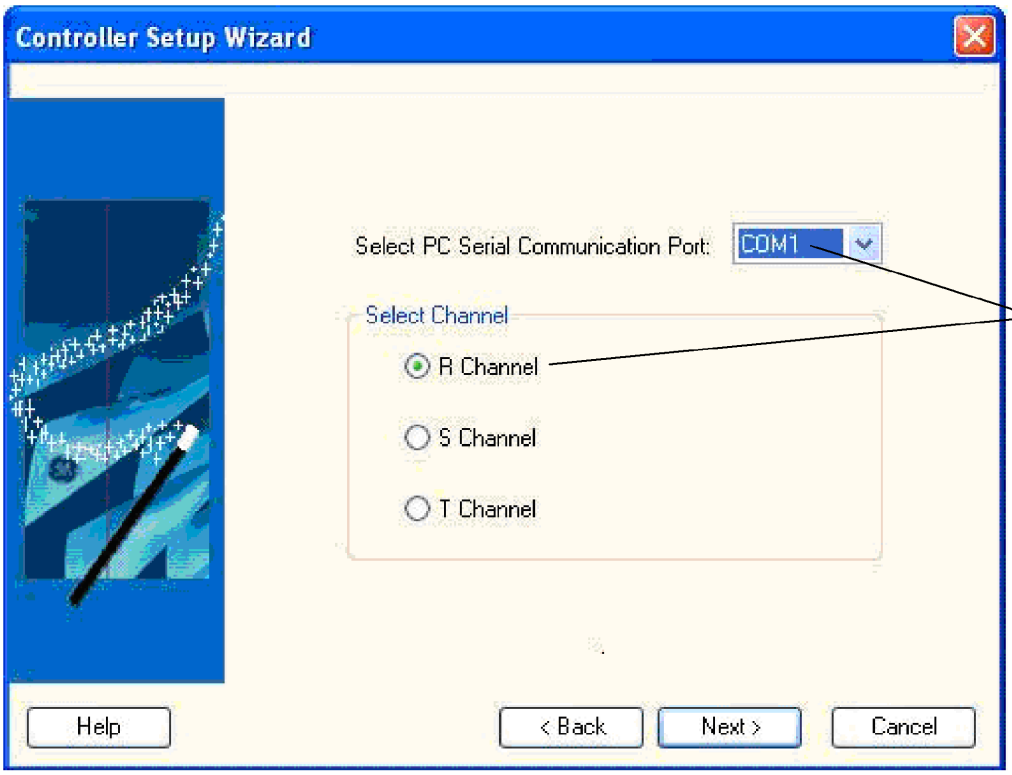
Mini-Serial Cable Adapter (Part #: 336A4929G1)

➢ **To set the controller IP address using a serial cable (recommended)**

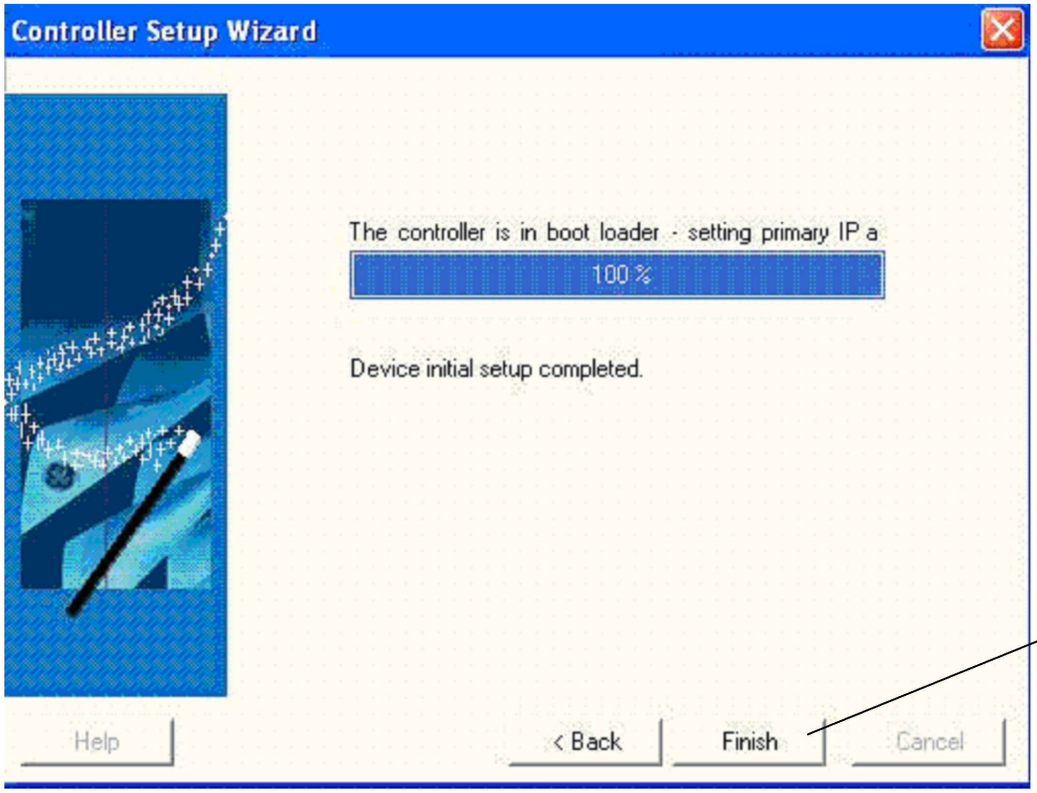
1. Connect the appropriate serial cables to the COM1 port of the R controller.
2. From the ToolboxST application, open the Mark VIe control *.tcw file.



3. From the **Device** menu, select **Download** and **Controller Setup**.
4. Select **Configure Network Address** and click **Next**.



From the **Select PC Serial Communication Port**: drop-down list, select the port, then select the channel, and click **Next**.



Click **Finish**.

5. Move the serial cable to the S controller.
6. Repeat steps 5 through 8 for the S controller.
7. Move the serial cable to the T controller.
8. Repeat steps 5 through 8 for the T controller.

Note Each controller is automatically restarted after the Controller Setup Wizard completes.

Using Flash Card Reader

Refer to GEI-100665, Mark VIe Controllers (UCCx and UCSx) Instruction Guide for the specifications to configure the IP address for each controller using CompactFlash®.

For non-UCSB controllers, a CompactFlash card reader sets the controller IP address by flashing the controllers.

Note If you are using Windows 7, you may have to run the ToolboxST application as an Administrator to format a CompactFlash card.



Flash Card Reader

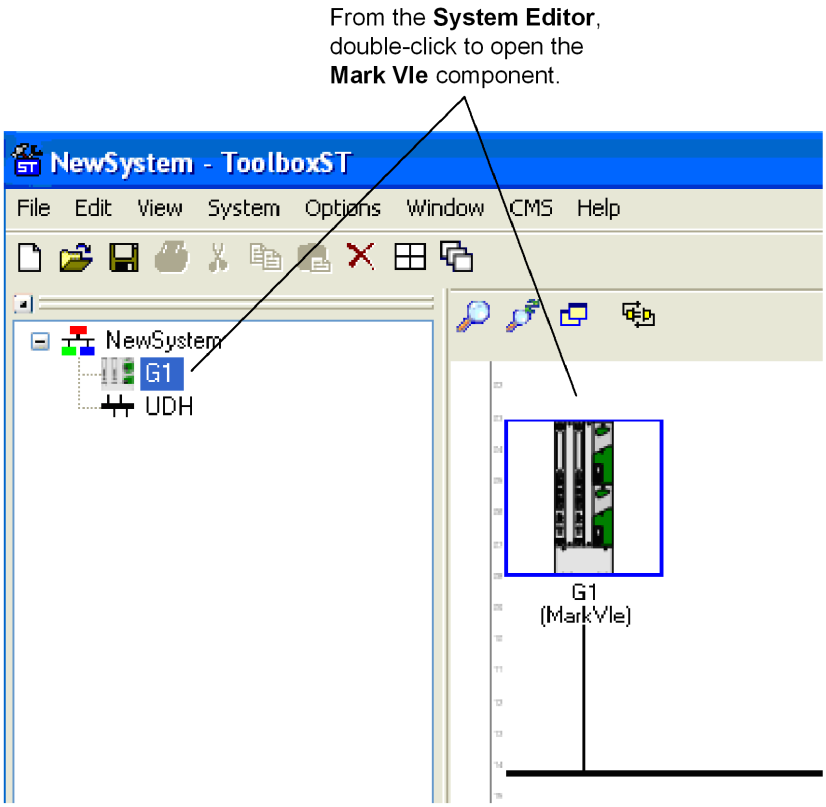
➢ **To set the controller IP address by flashing the controllers**

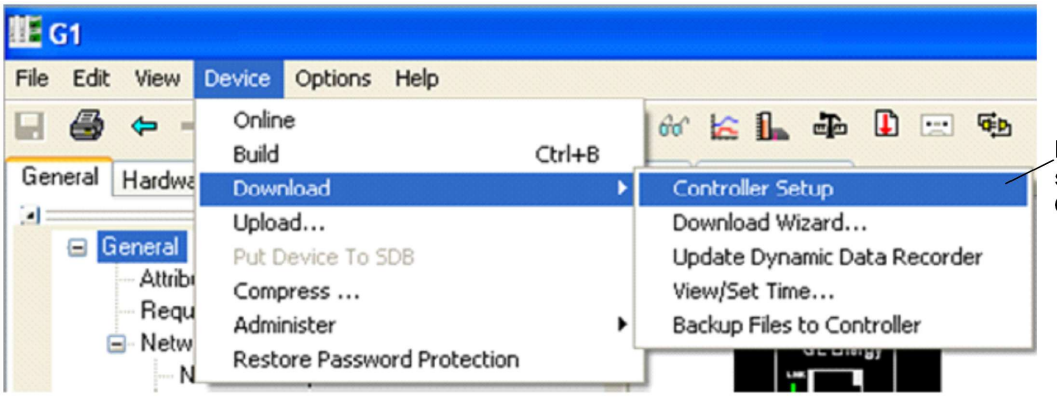
This procedure may be completed before power is applied to the control cabinet.

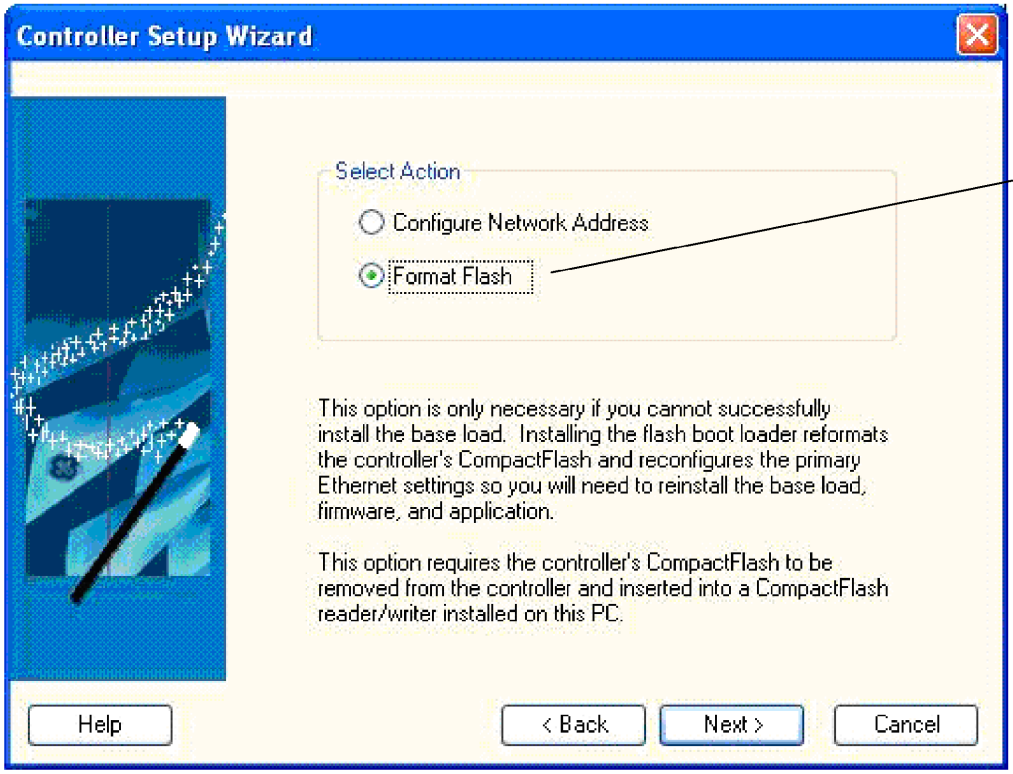
1. From the ToolboxST application, open the Mark VIe control .tcw file.
2. Turn OFF power to the controller and unplug the CompactFlash from the controller.



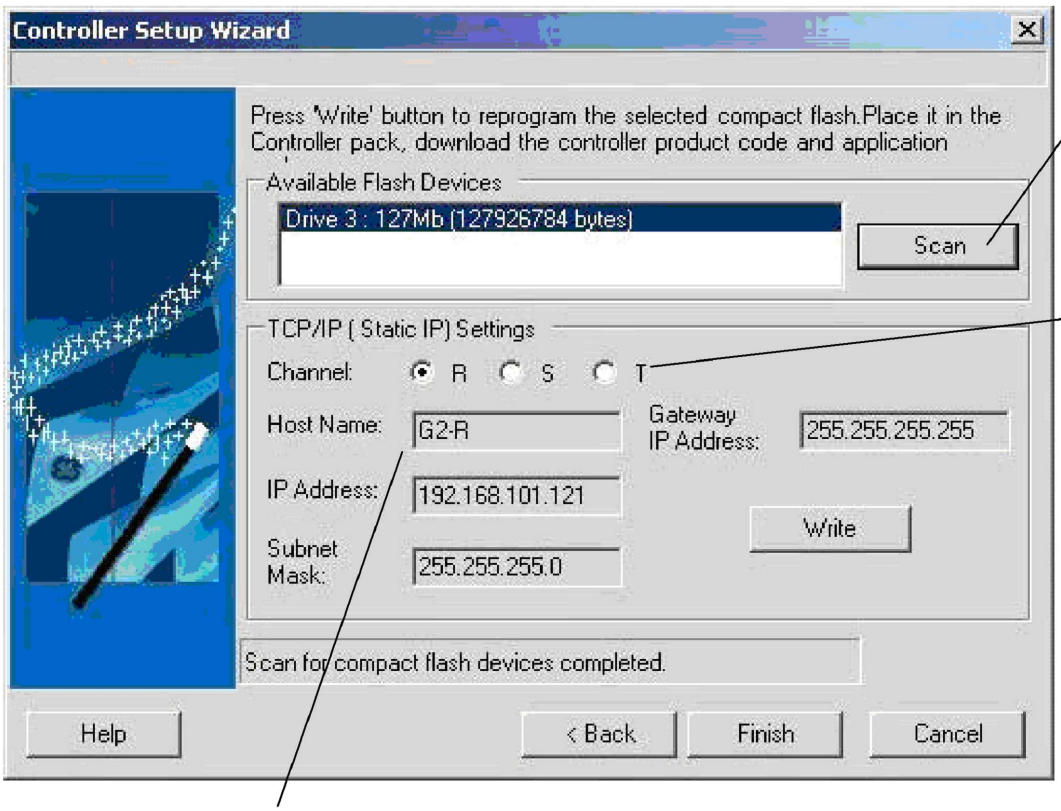
3. Insert the CompactFlash into the card reader, then plug the card reader into the USB port.







Select **Format Flash** and click **Next**.



Click **Scan**. The health of the CompactFlash is verified.

Select the controller (R, S, or T) for which the IP address is to be set.

Verify the correct IP address and controller name. The IP is 192.168.101.1xy (where x = controller number, 1 for G1, 2 for G, and such) and y = TMR number (1 for R, 2 for S, 3 for T).

Note If Scan cannot locate the flash card, either the card reader is improperly connected or the flash card is faulty.

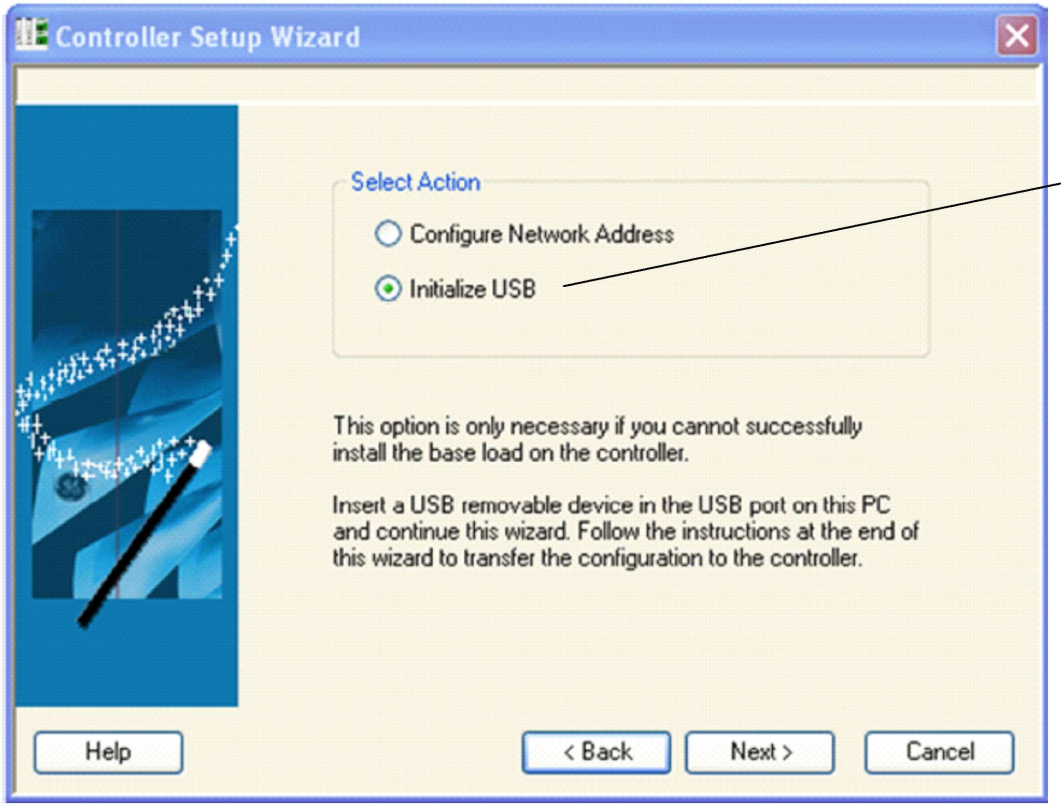
4. Click **Write** to format the CompactFlash.
5. Verify the **CompactFlash download completed successfully** message displays and click **Finish**.
6. Insert the CompactFlash into the controller.
7. Apply power to the controller (the red boot LED should flash at 1 Hz).
8. Repeat steps 2 through 7 for each controller.

Initialize USB

➢ **To assign an IP address to the UCSB using the Initialize USB option**

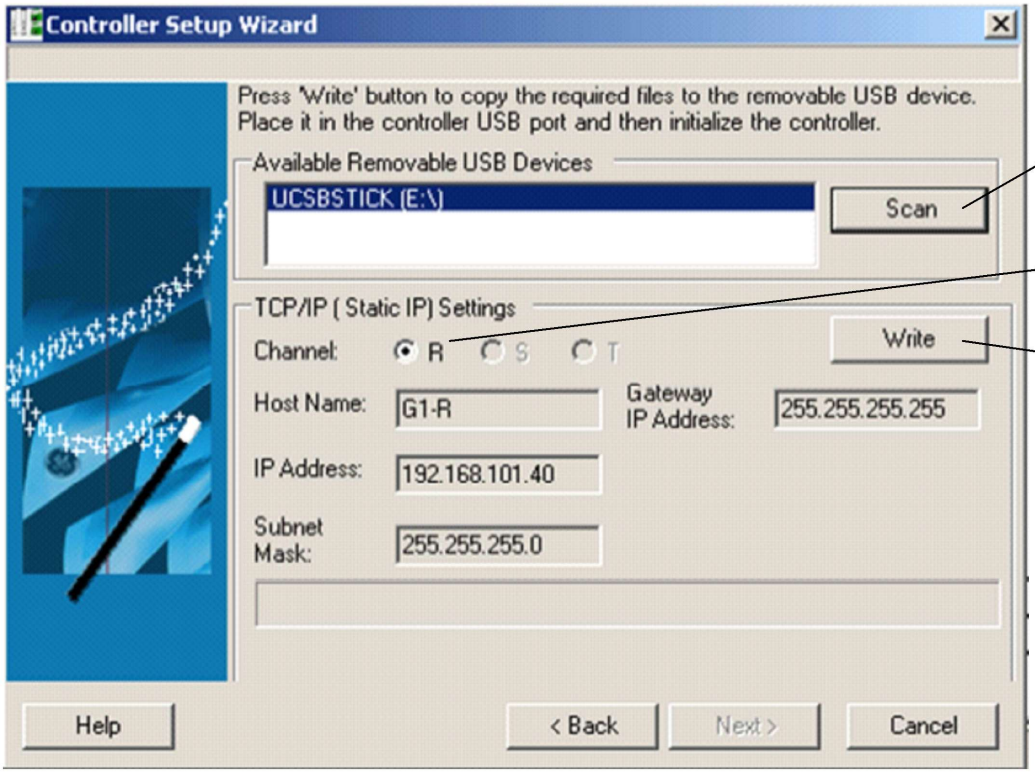
Note To assign the IP address using the Initialize USB option, a USB device is required. The USB device must be USB 2.0 compliant, non-encrypted, with a minimum capacity of 4 GB.

1. Insert the USB device into a USB port on the HMI.
2. From the project folder, open the **.tcw** file.
3. From the **ToolboxST System Editor**, open the **Component Editor** for the controller to be configured.
4. From the **Device** menu, select **Download** and **Controller Setup**. When the welcome wizard displays, click **Next**.



Select **Initialize USB**.

Click **Next**.



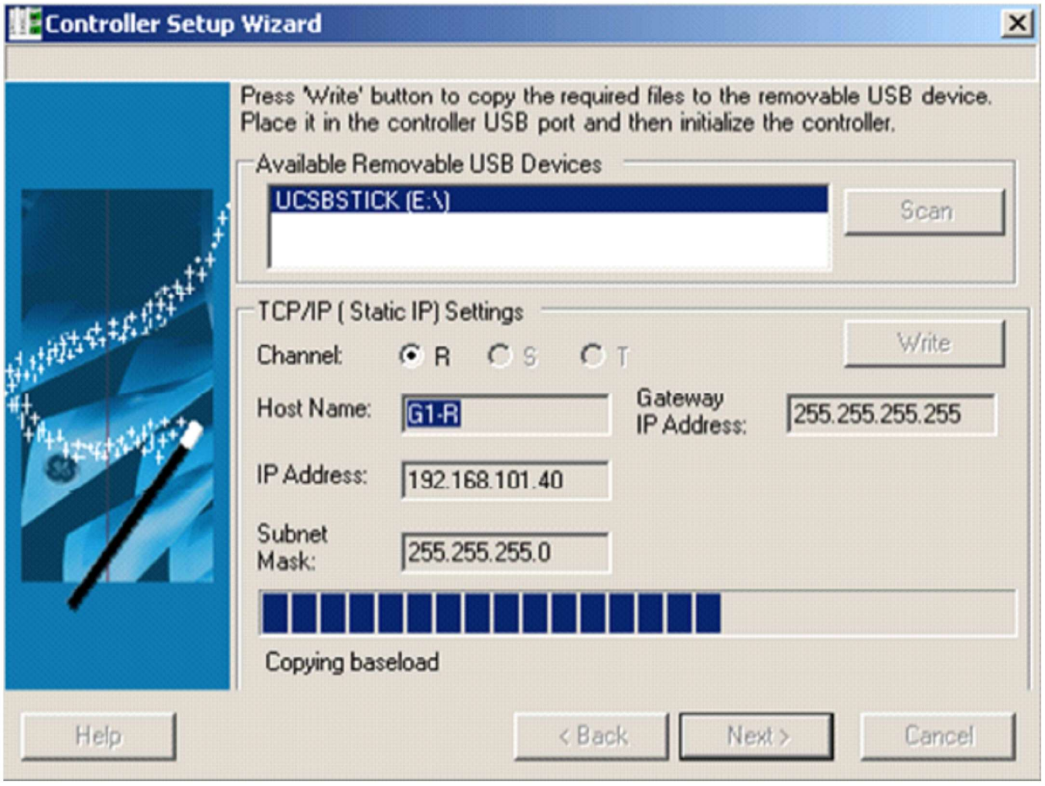
If the USB is not detected, click **Scan** to locate the USB.

From the **TCP/IP (Static IP) Settings** section, select **R**.

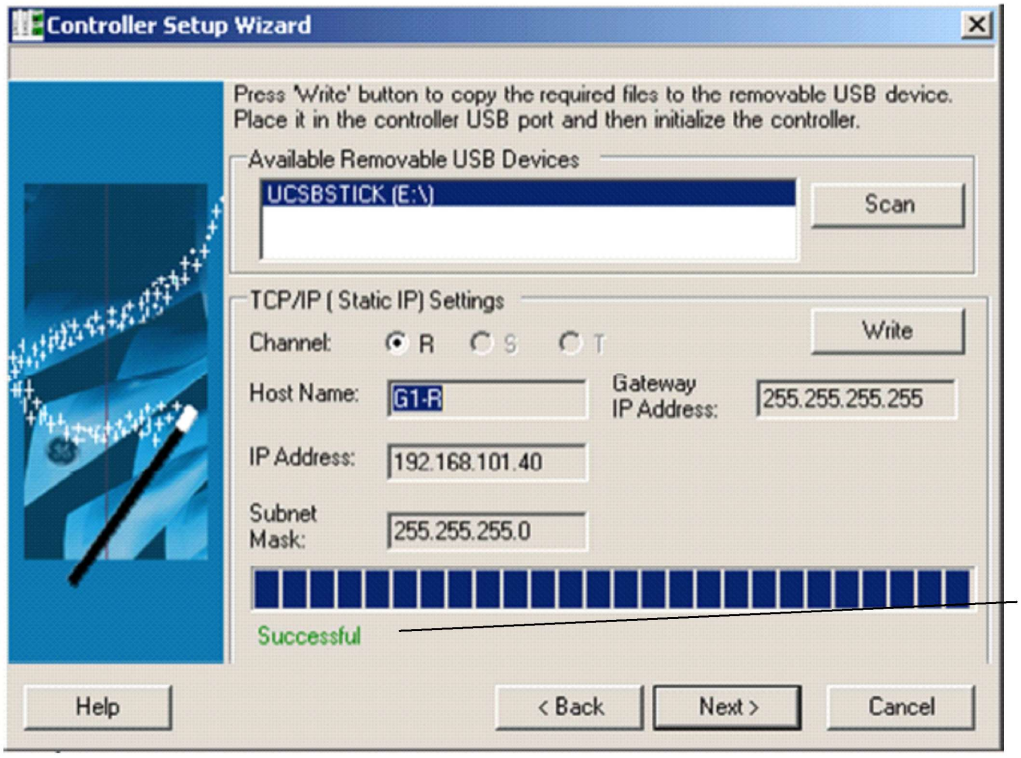
Click **Write**.

5. If the following message box displays, click **Yes**.



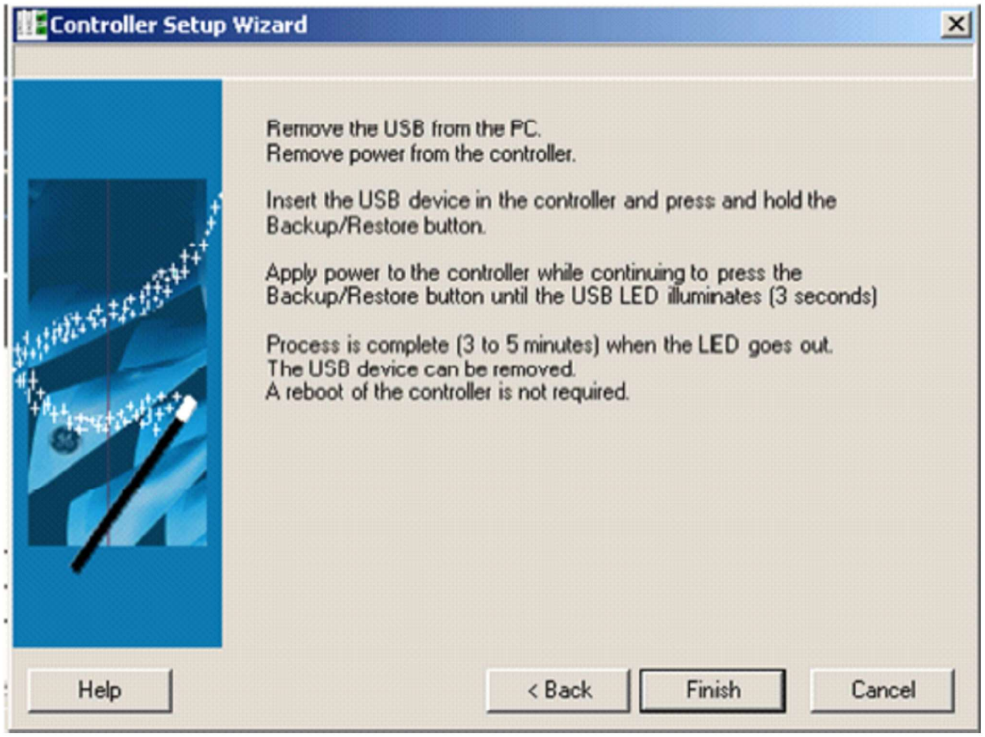


A progress bar displays. Wait until the writing process to the USB is completed.



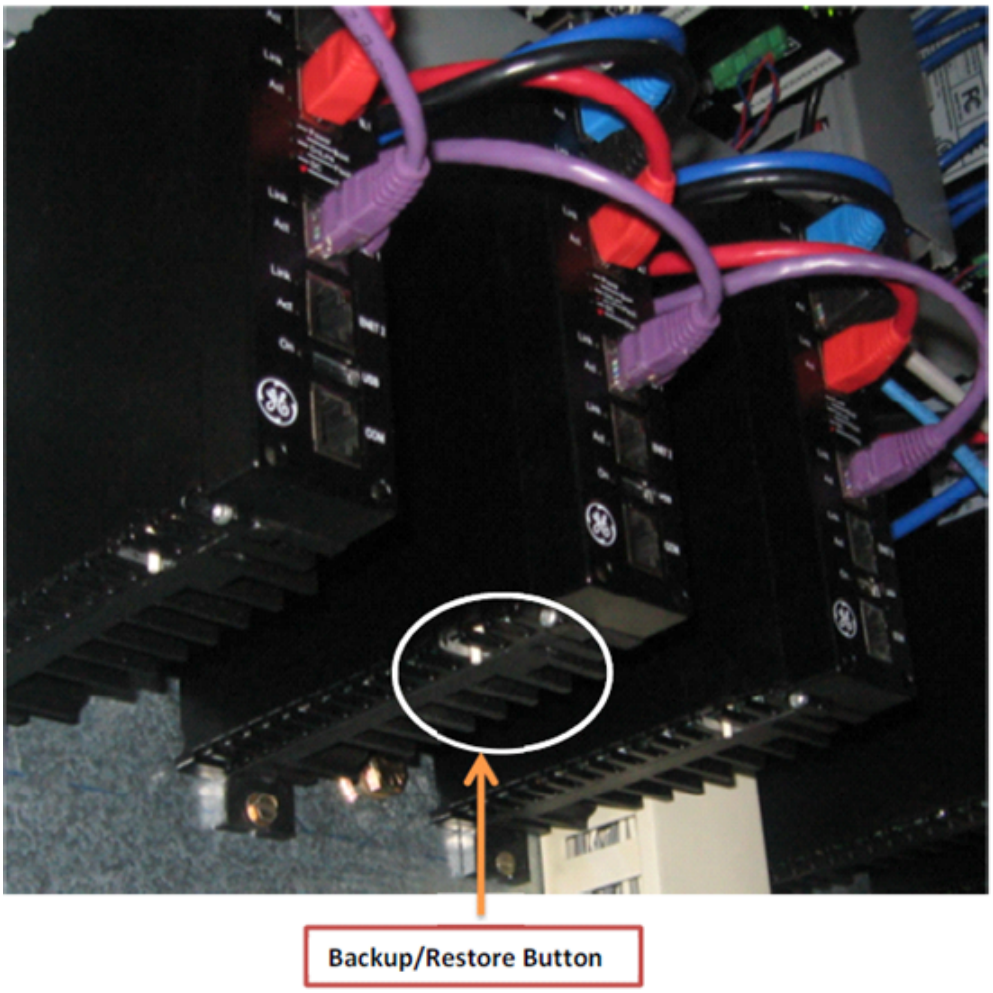
When **Successful** displays, click **Next**.

A notes screen displays.

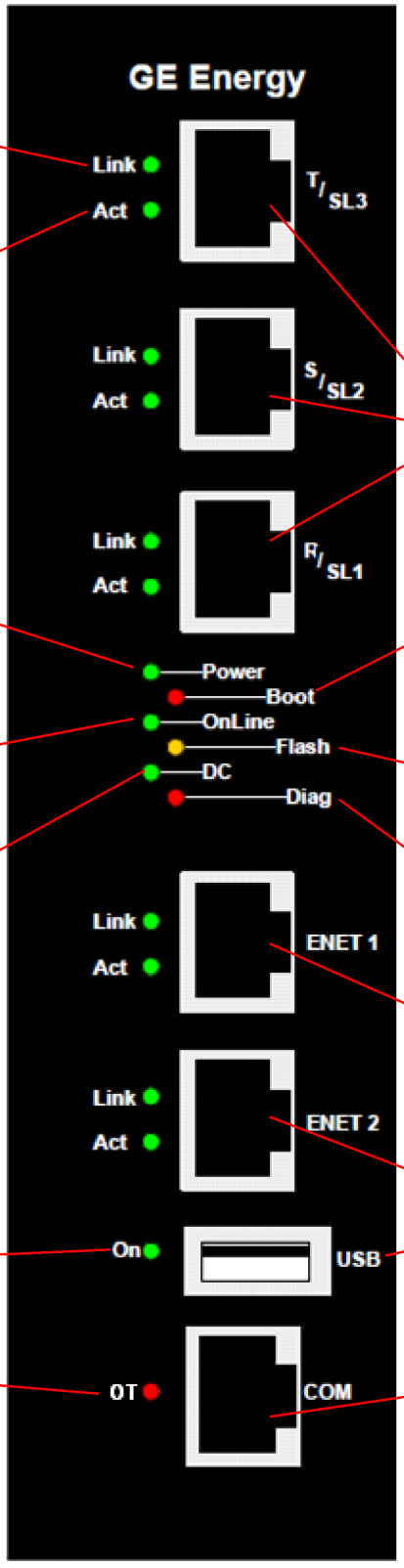


Read the notes carefully before clicking **Finish**. Steps mentioned in this window have to be performed using the USB device for restore/ Configuring the IP address for the UCSB controller.

Note The backup/restore button in the UCSB controller is on the bottom of the controller. (Refer to the following figure.)



Note If the USB LED flashes at a 1 Hz rate, a failure has occurred. Remove the USB drive or retry.



Link
Solid Green =
Ethernet PHY on the UCSB has established a link with an Ethernet switch port.

Act
Solid Green =
Packet traffic on an Ethernet interface. May flash if the traffic is low.

Ethernet connection for the R, S, and T I/O network.

Power
Solid Green =
Internal 5 V supply is up and regulating. UCSB converts the incoming 28 V dc to 5 V dc. All other internal power planes are derived from 5 V.

Boot
Solid Red =
Displays solid red during the boot process. Flashes when an error is detected.

OnLine
Solid Green =
Controller is online and running application code.

Flash
Blinks Amber =
Blinks when any flash device is being accessed.

DC
Solid Green =
This controller is the designated controller.

Diag
Solid Red =
Controller has a diagnostic available.

Ethernet connection for the Unit Data Highway (UDH) for communication with HMIs and other control equipment.

Optional **CDH**.

On
Solid Green =
USB is active.

USB connection.

OT
Over-temperature Status:
Yellow = An over-temperature alarm is present
Red = A temperature trip has occurred

COM1 RS232C
For initial controller setup.

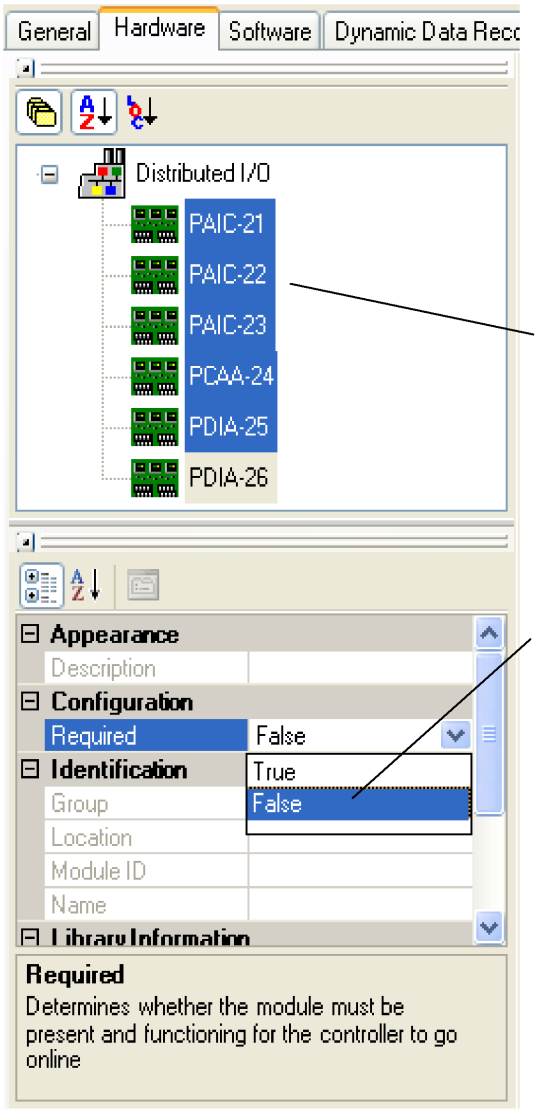
UCSB Front View

6. Repeat this procedure for the **S** and **T** controllers.

Downloading to the Controller

➢ **To download to the controller**

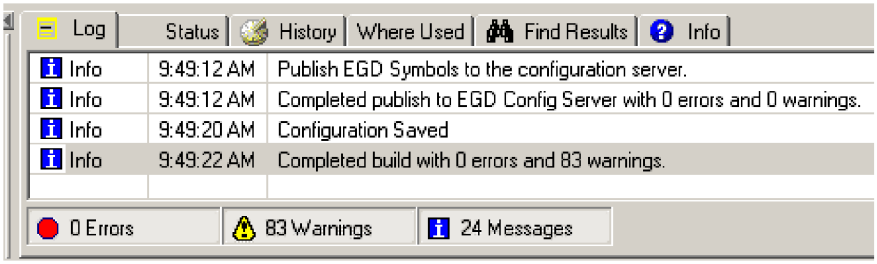
1. From the Mark VIe project folder, open the .tcw file.
2. From the **System Editor**, double-click the Mark VIe Controller. The **Component Editor** displays.



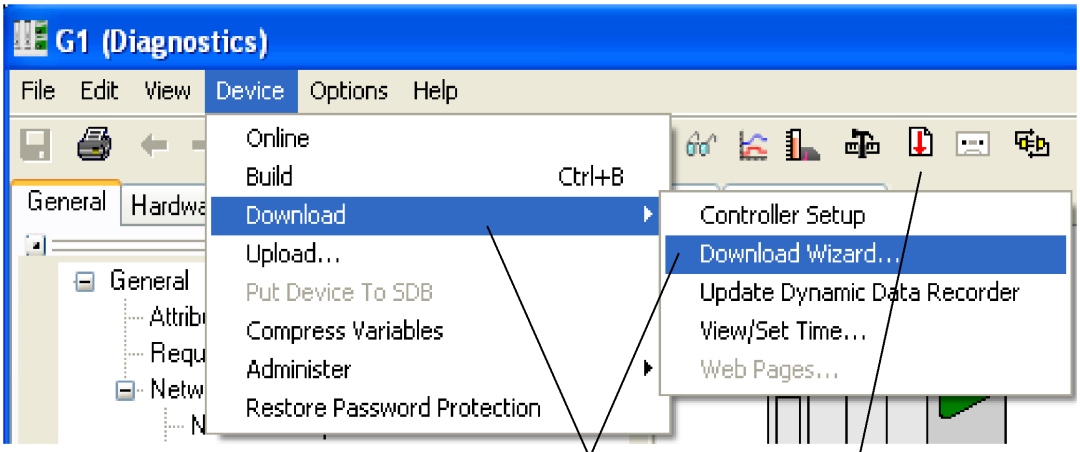
From the **Component Editor Hardware** tab, select all packs.

From the **Property Editor**, select **False** as the **Configuration Required** property.

3. Perform the Build command to the controller, then from the **Component InfoView**, verify that there are no errors.



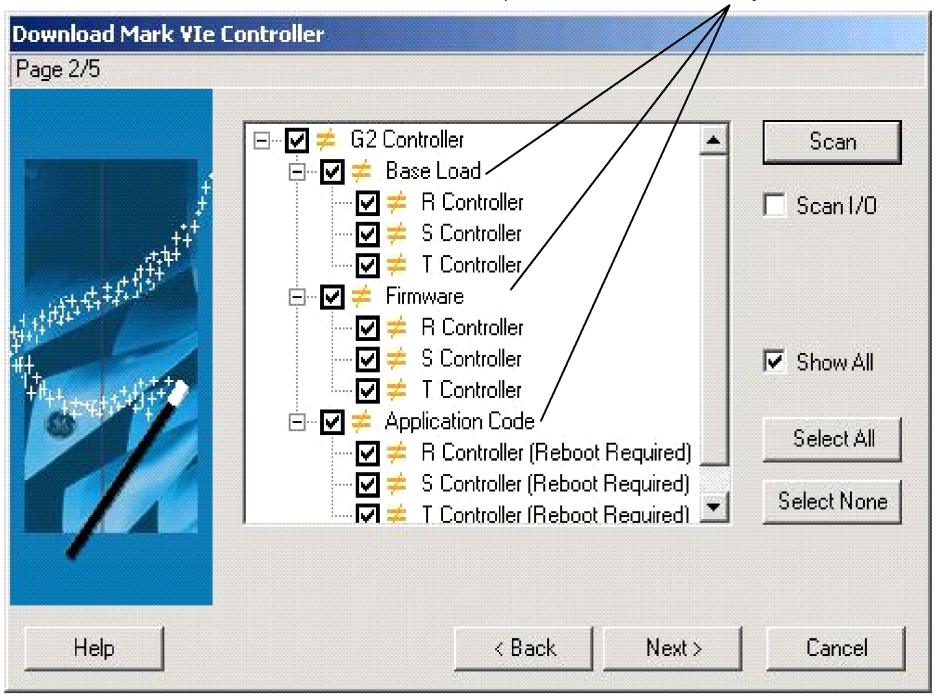
Note In case of Build errors, troubleshoot the issue before proceeding. Refer to *GEH-6721, Volumes I, II, and III for additional information*.



From the **Device** menu, select **Download**, then **Download Wizard…** or click the **Download** icon.

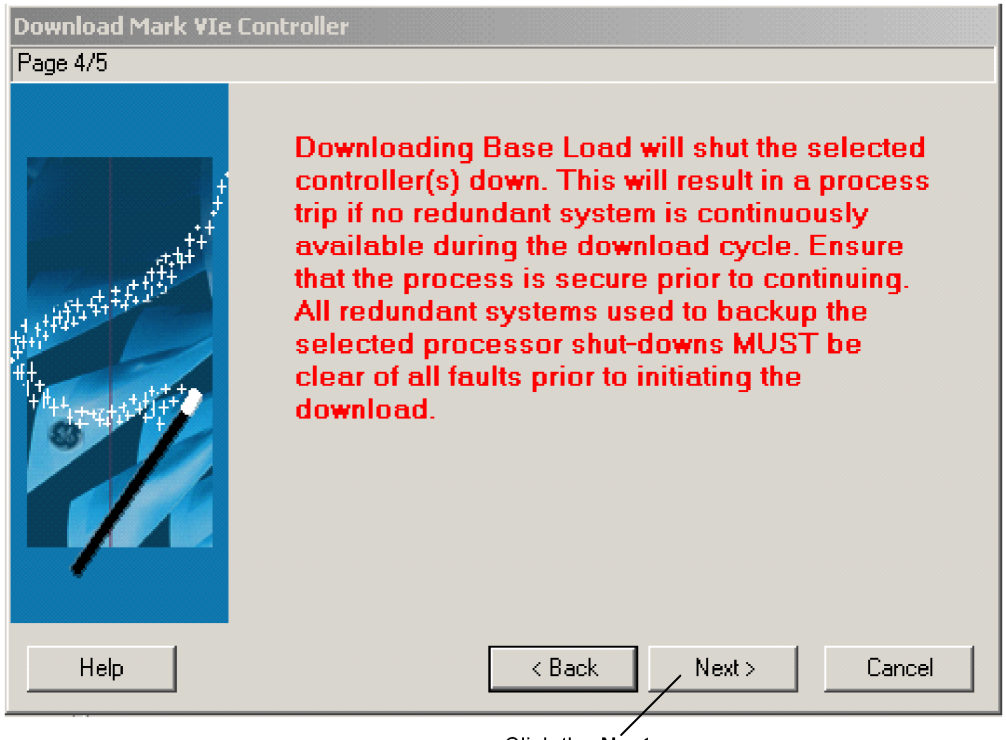
The **Controller Setup Wizard** displays.

Verify that the **Base Load**, **Firmware**, and **Application Code** check boxes are selected for all controllers (for **Dual** and **TMR** devices). Select controllers only at this time.



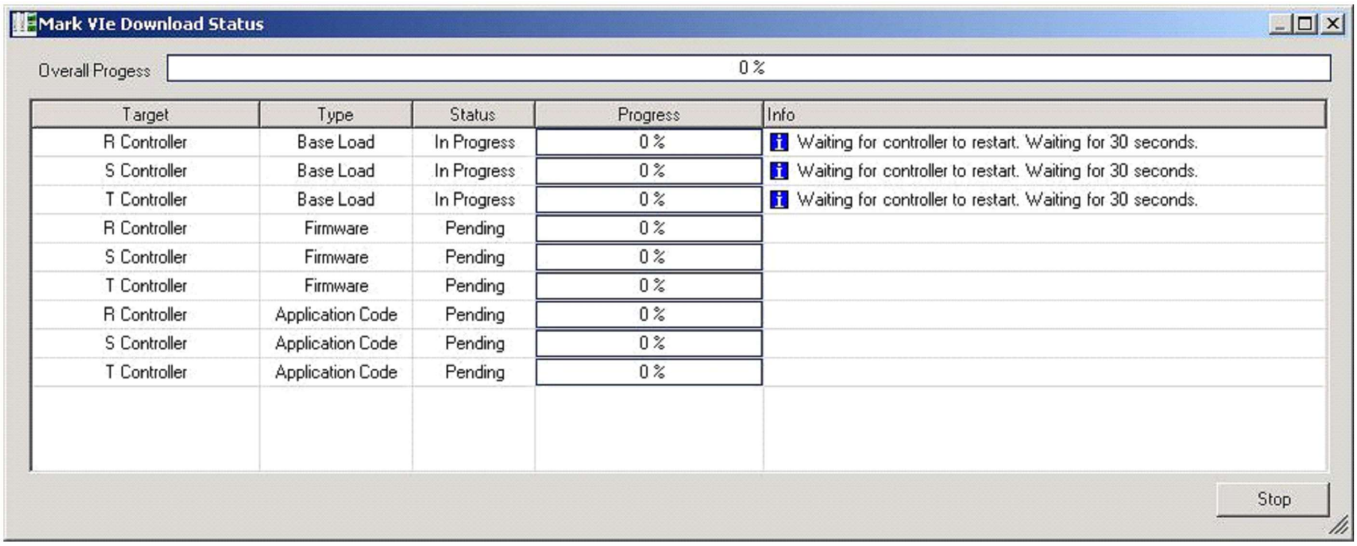
Note The base load and/ or firmware might indicate Green Equal if they have already been downloaded earlier during FAT or factory testing. It is recommended that the respective check boxes are manually selected during first time download.

The following message displays:

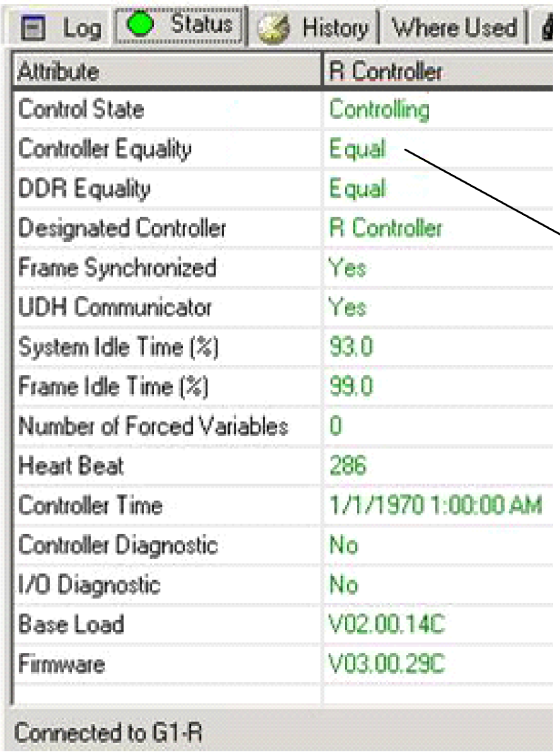


Click the **Next** button to continue.

4. The controller restarts itself at each step after base load, firmware, and application code during the download.



5. When the download is complete, wait for two minutes then go online with the controller and verify the controller equality for all three controllers.

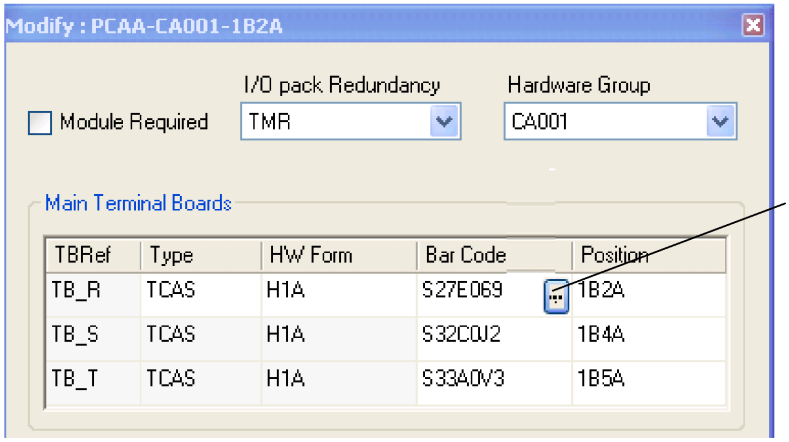


The **Controller Equality** displays in the **Status** tab.

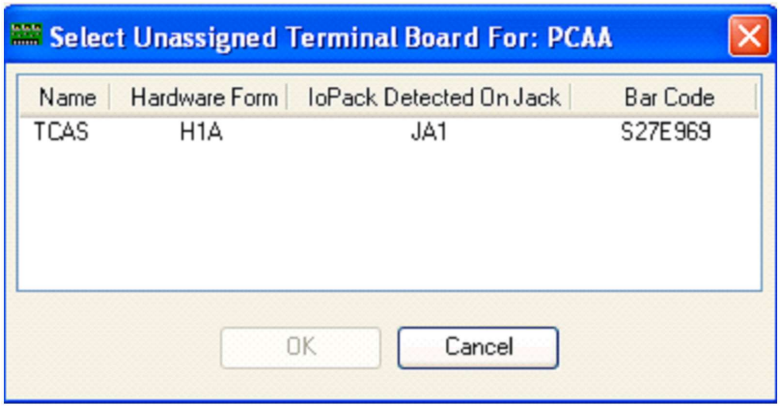
Downloading to the I/O Packs

➢ **To download to the I/O packs**

1. From the Mark VIe project folder, open the .tcw file.
2. From the **ToolboxST System Editor**, open the **G1** component.
3. From the **Hardware** tab, right-click a pack and select **Modify**.
4. Enter the bar codes (found on terminal board stickers) in the I/O pack configuration.
5. Cross-verify the entered bar codes with the Dallas ID read by the controller as follows.



Go online with the controller, then from the **Modify:** dialog box, click the ellipsis button to enter the bar code.



   a. Verify that the entered bar code number is displayed in the unassigned bar code list. It is also possible to select the bar code number from the unassigned list and click **OK**.

   b. Verify each bar code number for all packs.

6. Build the software and verify that there are no errors.
7. From the **Device** menu, select **Download**, **Download Wizard** and click **Next**.
8. Select the **Scan I/O** box and the **Show All** check boxes.
9. Click the **Scan** button.

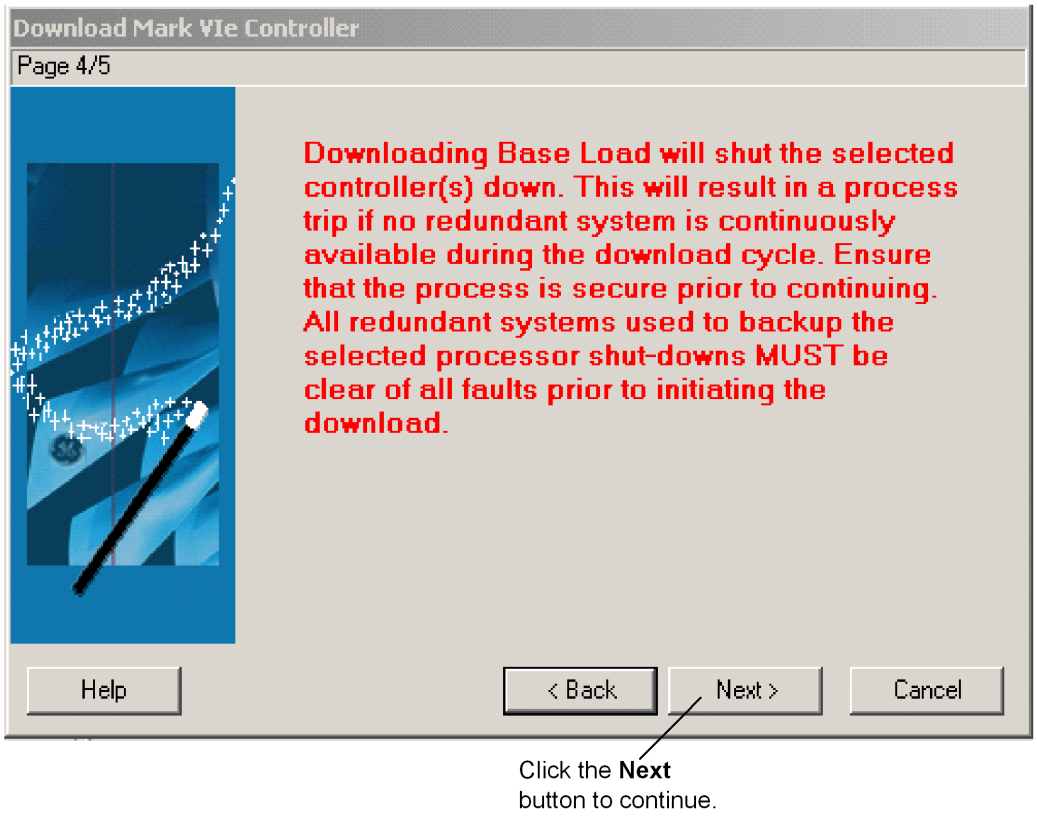
Note When opening the Download Wizard, the configured hardware is automatically scanned for healthiness if the scan was previously enabled.

10. Verify that the **Base load**, **Firmware** and **Parameters** check boxes are selected for all I/O packs.

Note Some I/O packs or even the entire cabinet might need to be restarted if some I/O packs are not being read or located by the scan. Rescan the hardware more than once, if necessary.

Note The base load and/ or firmware might indicate Green Equal status on some I/O packs if they were downloaded to the packs earlier during FAT or factory testing. Manually select the respective check boxes during the first-time download.

11. Select the **Download Backup File** and click Next. The following message displays.

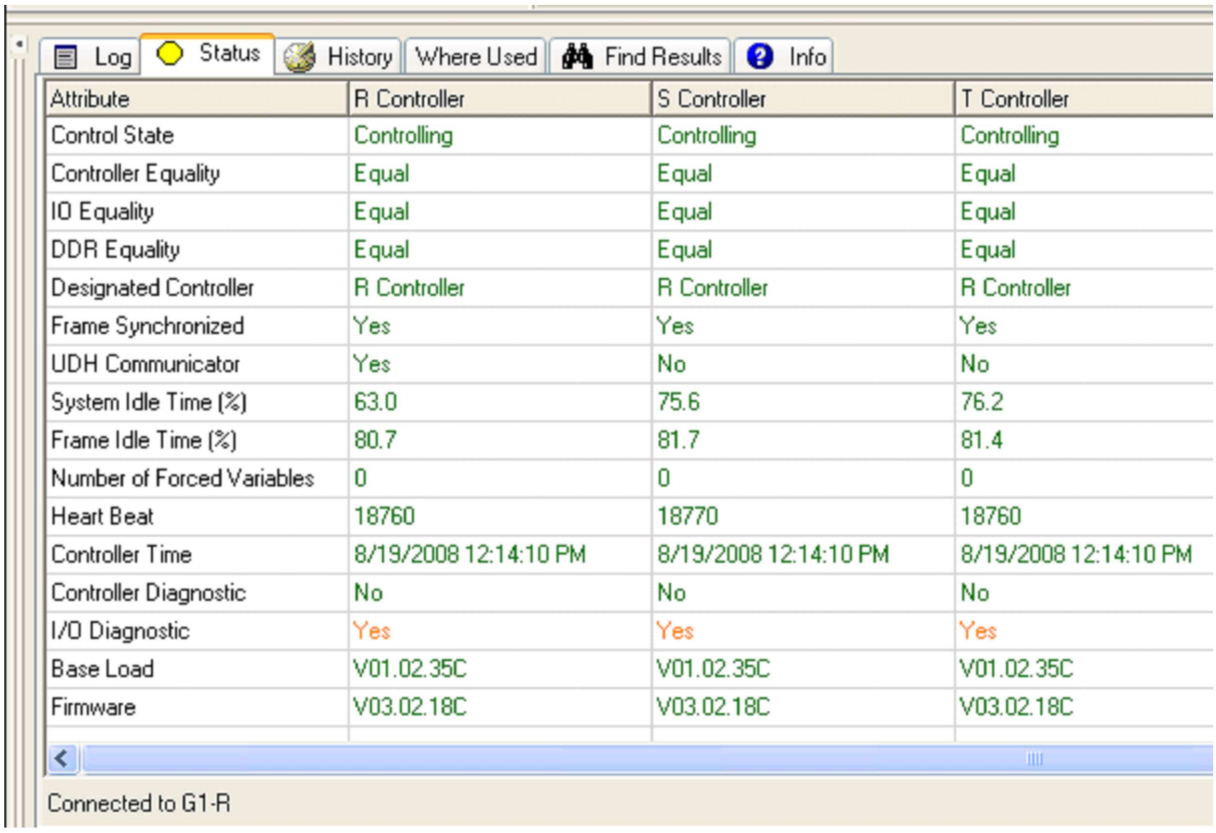


12. When the Download is complete, wait two minutes then go online and verify the status of the I/O packs.

13. To verify that the Download was successful for all the I/O packs, open the Download Wizard again and scan the hardware. The scan result should indicate all the controllers and I/O packs as Green Equal.

14. Verify the controller diagnostic alarms and reset alarms as follows:

    a. From the **Component InfoView**, select the **Status** tab.



Attribute	R Controller	S Controller	T Controller
Control State	Controlling	Controlling	Controlling
Controller Equality	Equal	Equal	Equal
IO Equality	Equal	Equal	Equal
DDR Equality	Equal	Equal	Equal
Designated Controller	R Controller	R Controller	R Controller
Frame Synchronized	Yes	Yes	Yes
UDH Communicator	Yes	No	No
System Idle Time (%)	63.0	75.6	76.2
Frame Idle Time (%)	80.7	81.7	81.4
Number of Forced Variables	0	0	0
Heart Beat	18760	18770	18760
Controller Time	8/19/2008 12:14:10 PM	8/19/2008 12:14:10 PM	8/19/2008 12:14:10 PM
Controller Diagnostic	No	No	No
I/O Diagnostic	Yes	Yes	Yes
Base Load	V01.02.35C	V01.02.35C	V01.02.35C
Firmware	V03.02.18C	V03.02.18C	V03.02.18C

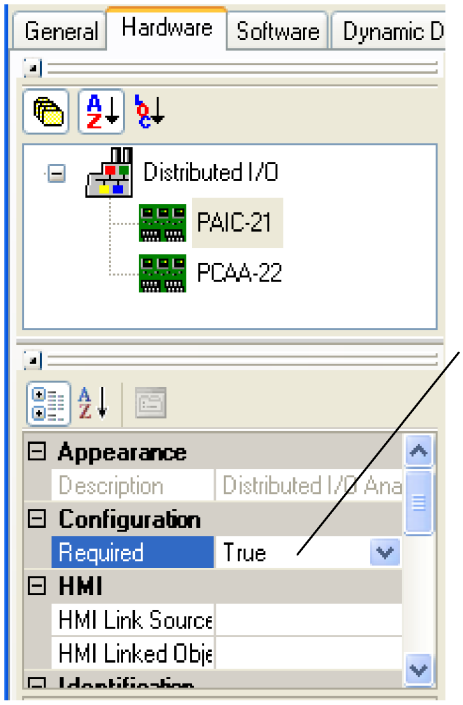
    b. Double-click the **Controller Diagnostic** status and select the **R** controller. Click the **Reset Fault** button in the diagnostic alarm window. Repeat for the S and T controllers.

15. Verify the I/O pack diagnostic alarms and reset alarms as follows:

    a. From the **Component InfoView**, select the **Status** tab.

    b. Double-click the **I/O Diagnostic** status and select **All** I/O packs from the drop-down list. Click the **Reset Alarms** button.

16. When the download is complete, select the **Hardware** tab and all I/O packs.



From the **Property Editor**, **Configuration Required** property, select **True**.

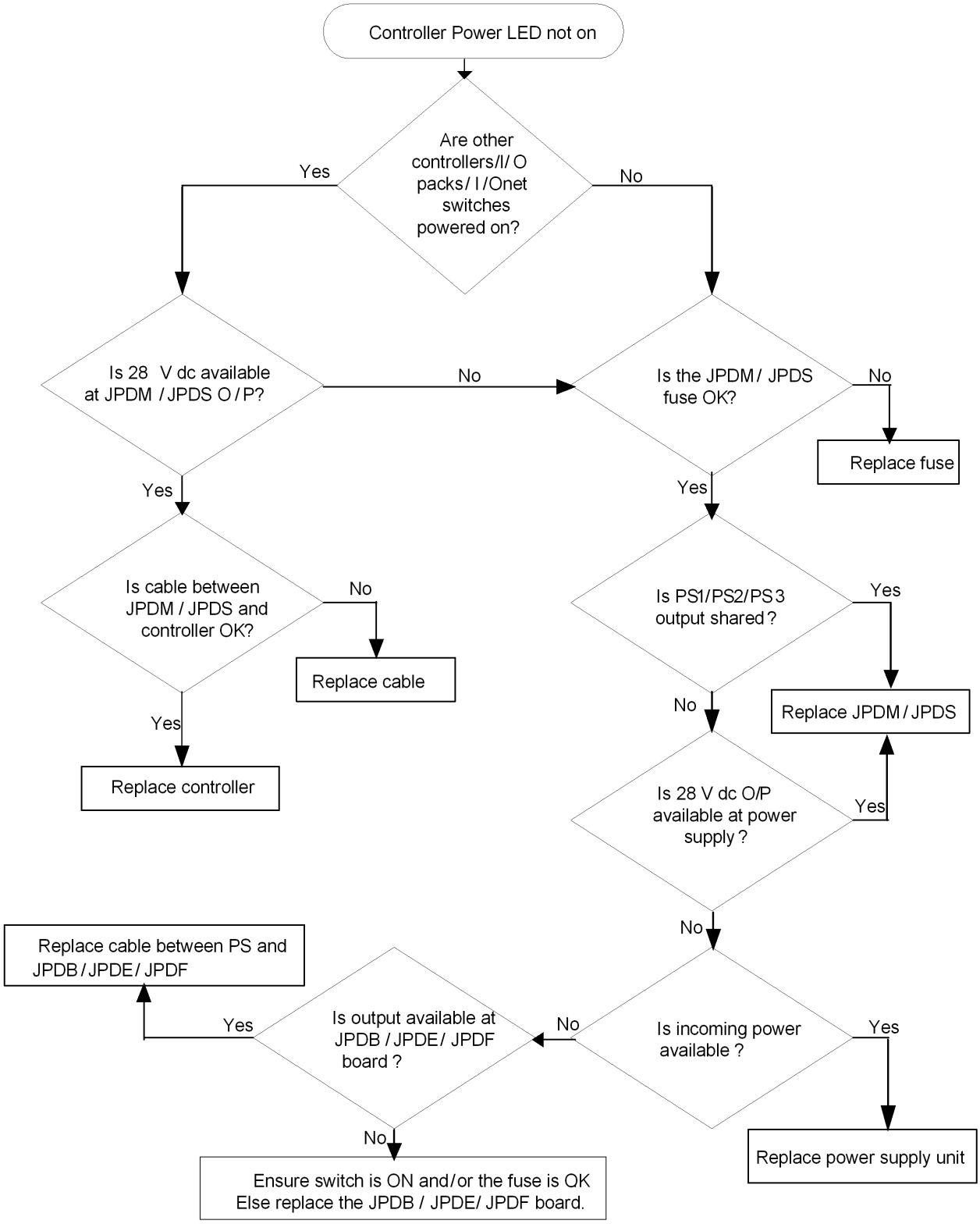
17. Perform the Build and Download commands to the application code for the controllers.

Troubleshooting

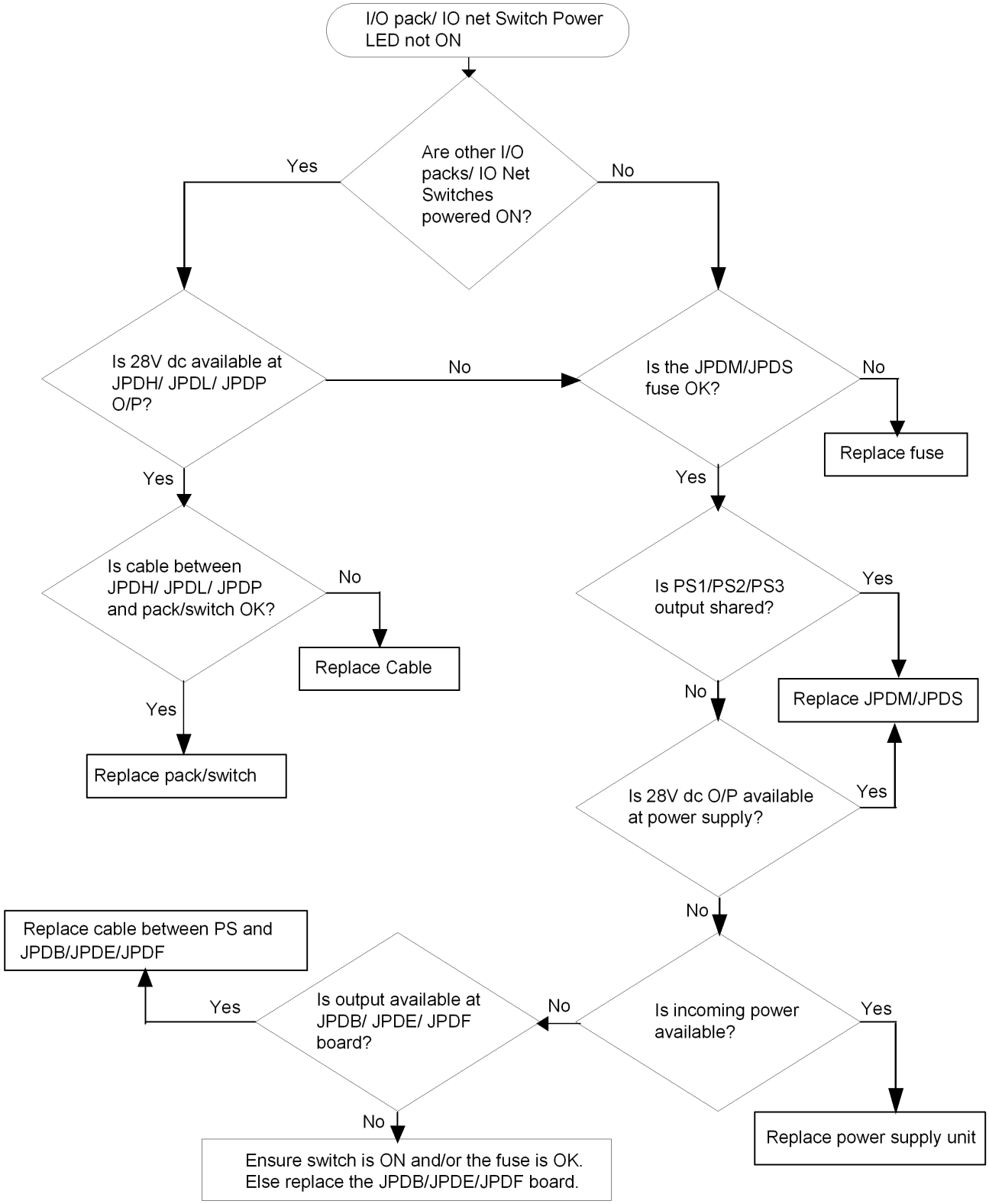
Possible troubleshooting issues include the following:

- *Controller is not turning on*
- *I/O packs / IONet switch are not turning on*
- *Unable to download to the controller*
- *Controller is not attaining a controlling state*
- *Unable to download to I/O packs*
- *Unable to communicate with an I/O pack*
- *Unable to download to the controller or I/O pack*
- *Communication loss between the controller and I/O pack*

Controller Is Not Turning On

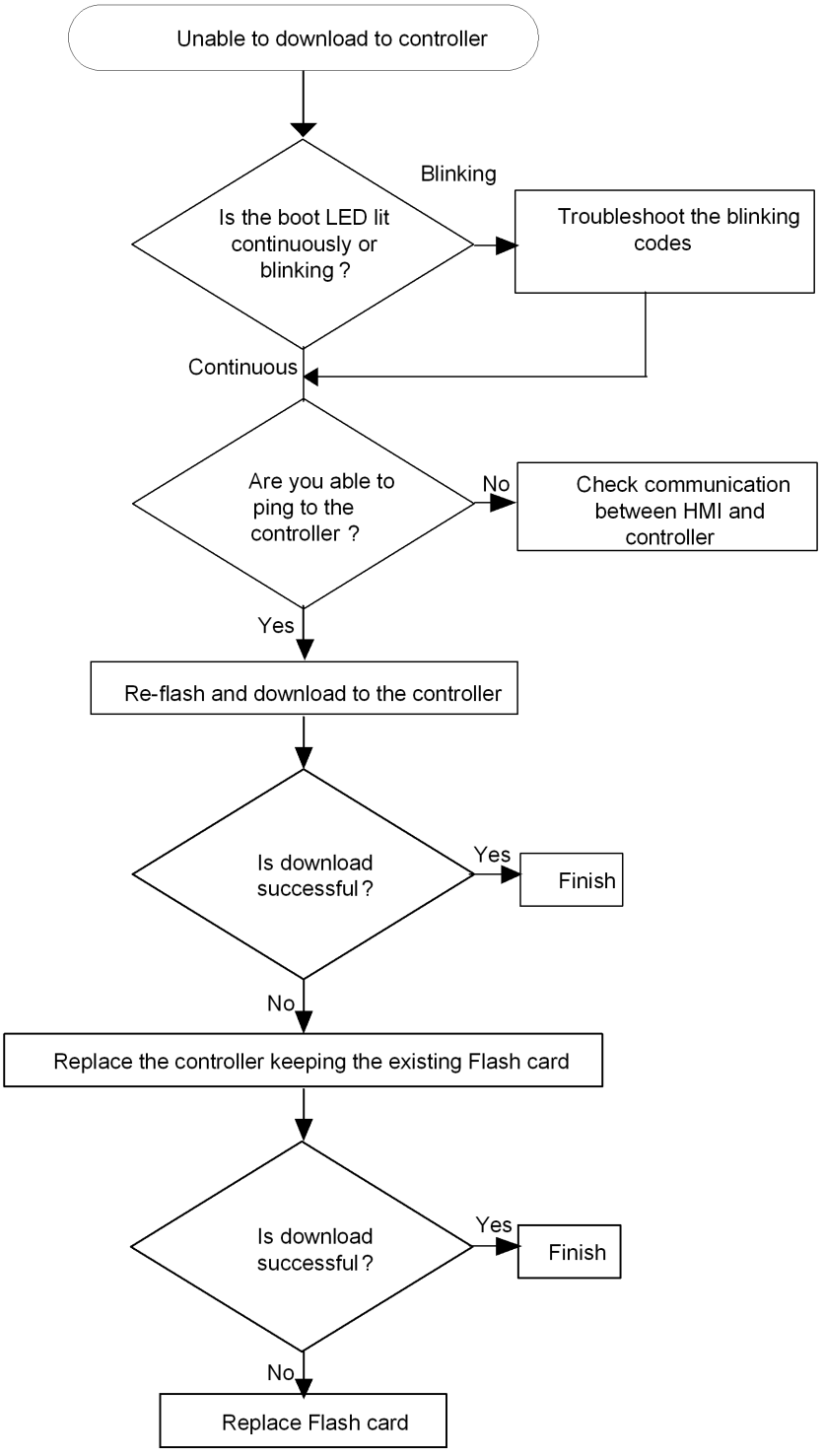


I/O Packs / IONet Switch Are Not Turning On

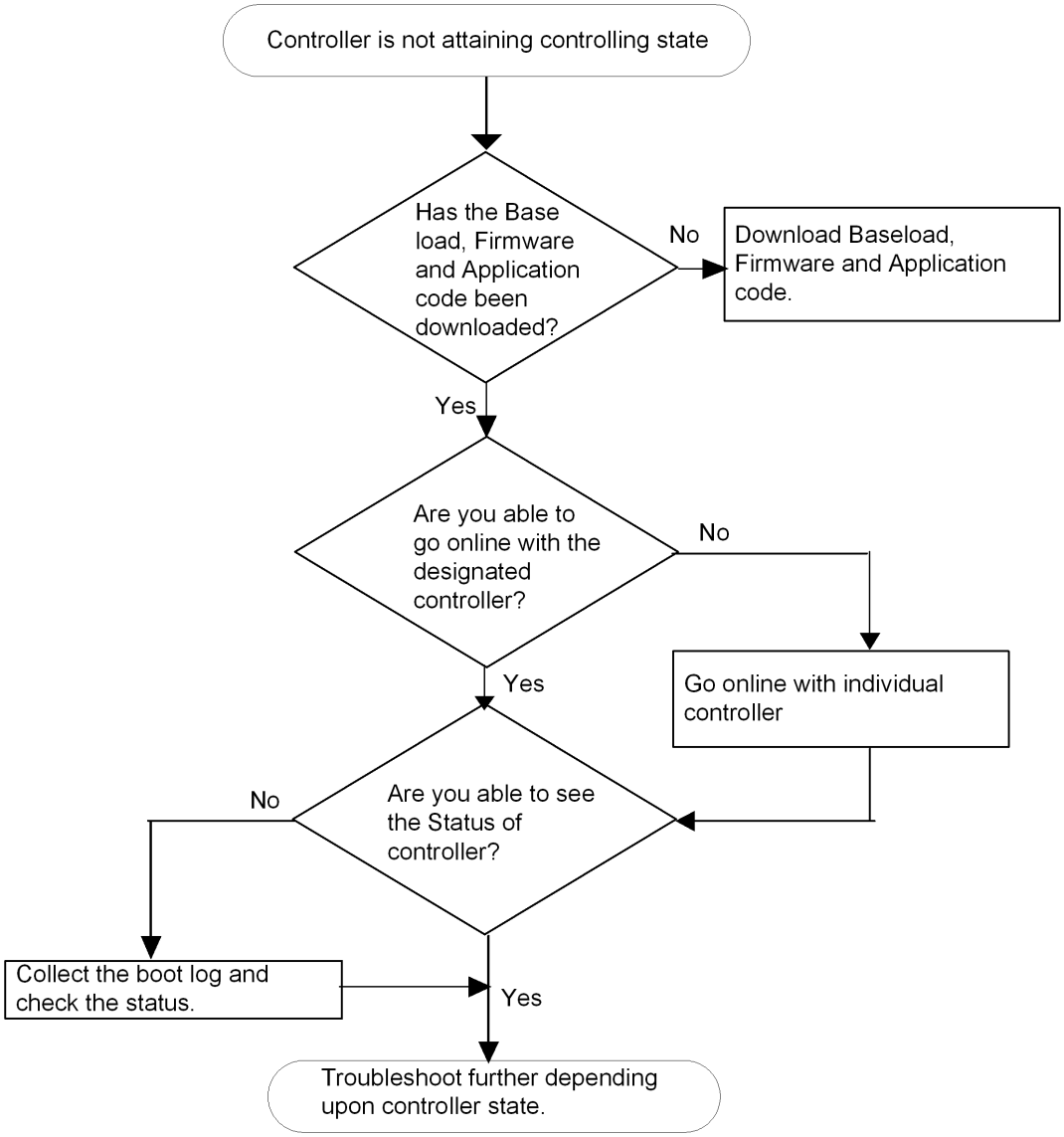


Unable to Download to the Controller

Before selecting the Download command, assign an IP address to the controller.



Controller is Not Attaining a Controlling State



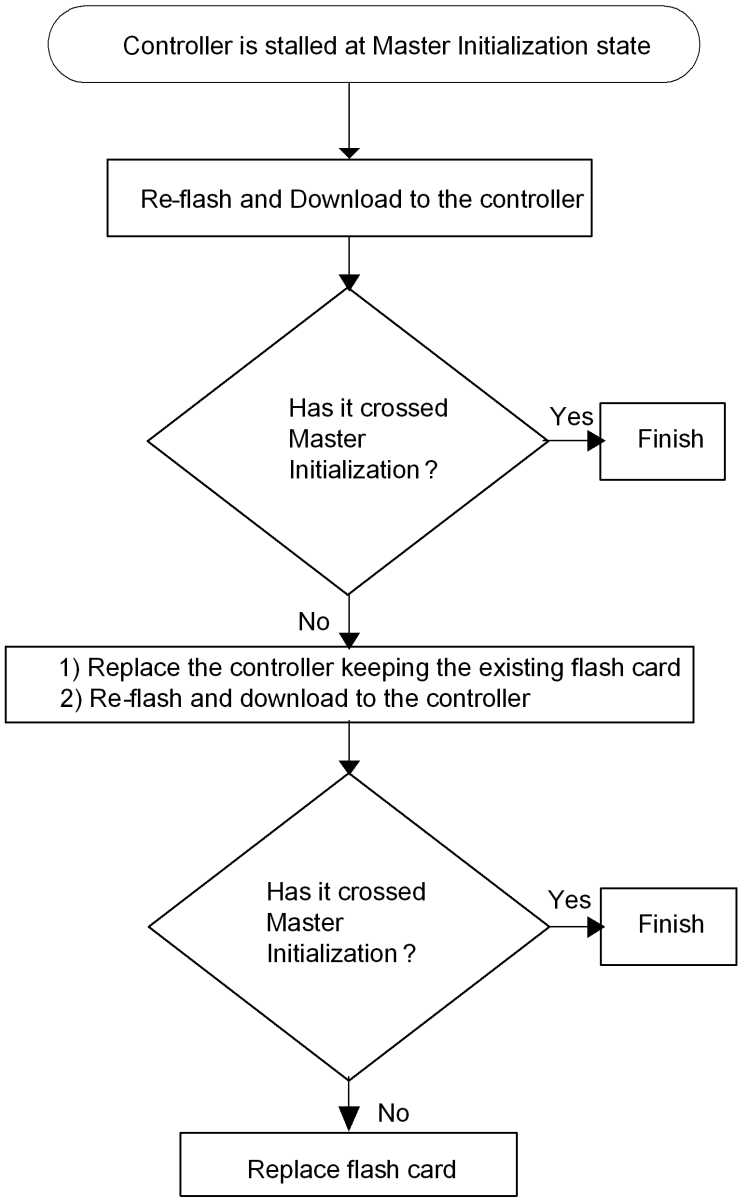
Controller States

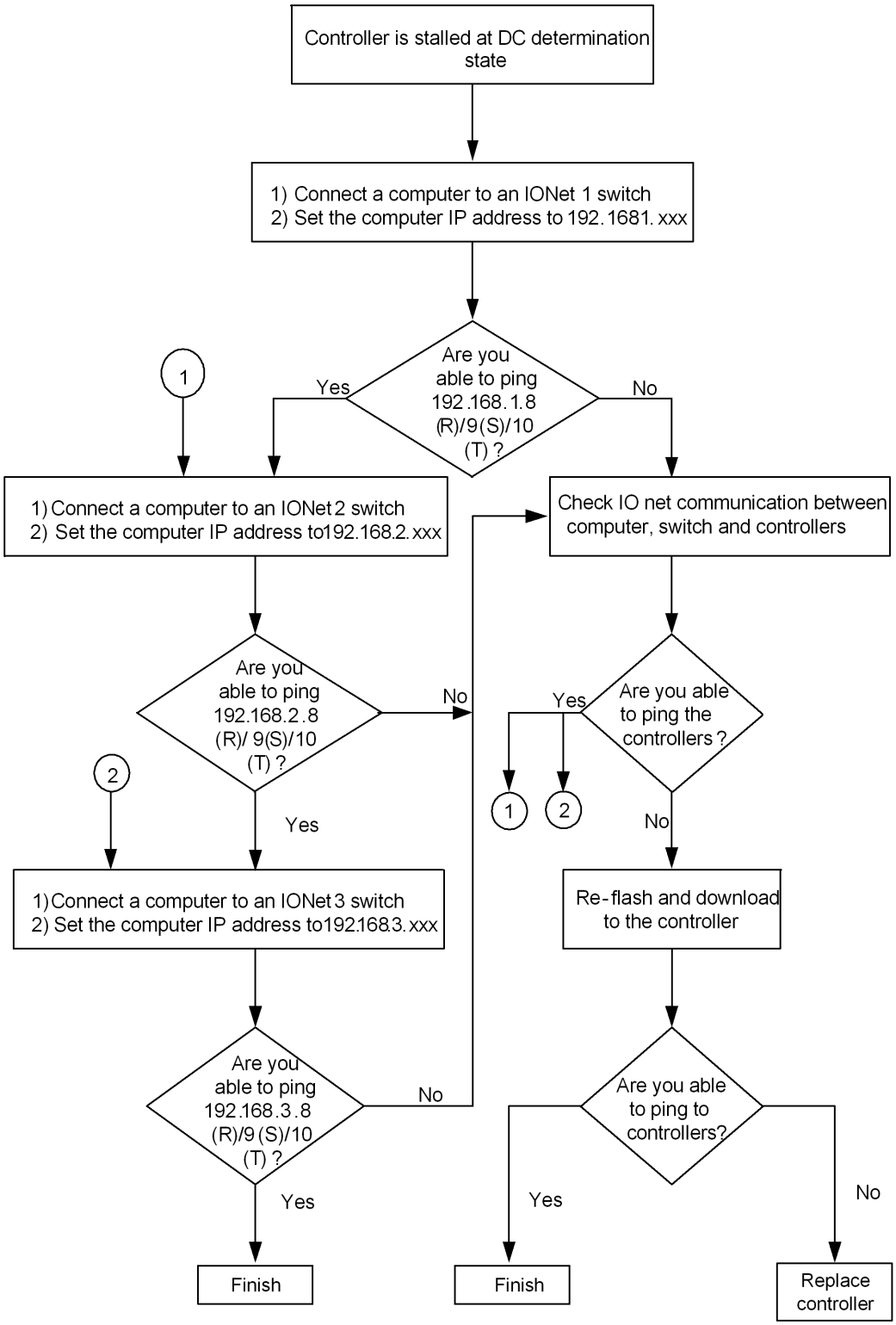
- Power-up
- d0-MASTER_INITIALIZATION
- d1-DC_DETERMINATION
- d2-DATA_INITIALIZATION
- d3-INPUTS_ENABLED
- d5-EXCH_INITIALIZATION
- d6-EXCHANGING
- d7-SEQUENCING
- d8-STANDBY
- dA-CONTROLLING

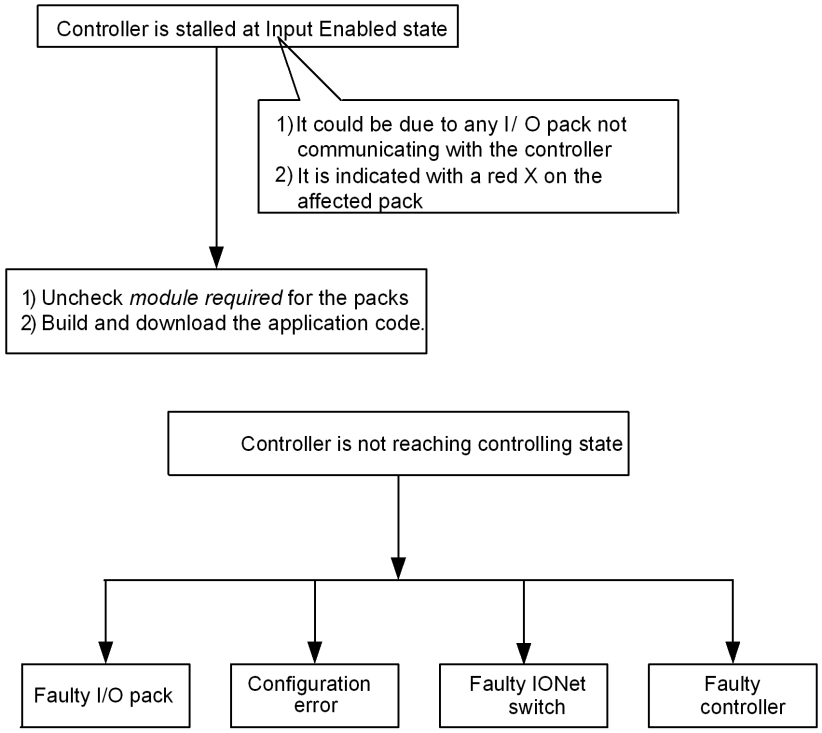
Pre-requisites for attaining a controlling state include the following.

- Controller should have the base load, firmware, and application code downloaded.
- All controllers (in dual or TMR) should communicate with each other over the IONet.
- All I/O packs (if configured as required) should be able to communicate with the controllers.

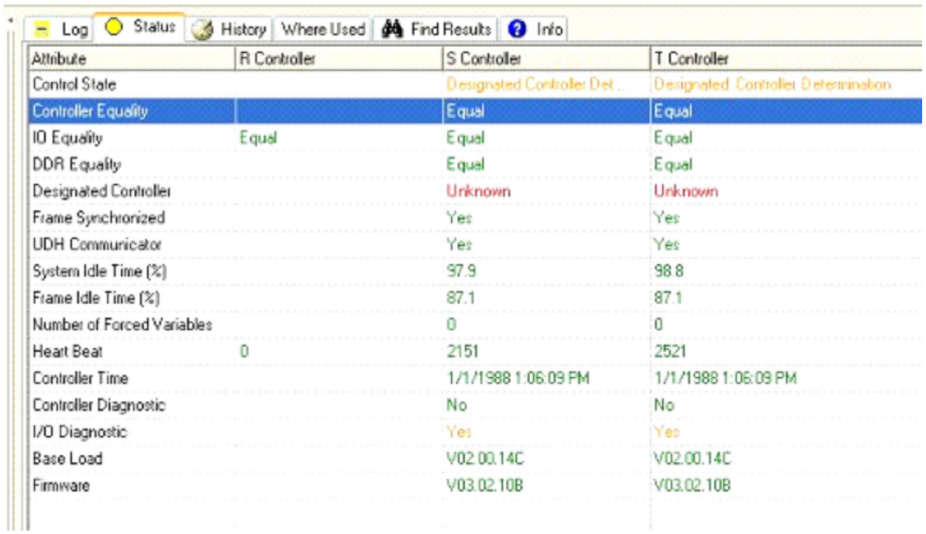
The Mark VIe system may not reach a controlling state due to faults in either the I/O pack, the Ethernet switch, or the controller itself. A single faulty controller can prevent the other two controllers from reaching a controlling state.




Controller is stalled at DC determination state
1) Connect a computer to an IONet 1 switch
2) Set the computer IP address to 192. 1681. xxx
Are you able to ping 192 .168. 1. 8 (R)/9(S)/10 (T) ?
Yes
No
1
1) Connect a computer to an IONet 2 switch
2) Set the computer IP address to192.168.2. xxx
Check IO net communication between computer, switch and controllers
Are you able to ping 192.168.2 .8 (R)/ 9(S)/10 (T) ?
No
Yes
Are you able to ping the controllers ?
2
Yes
1
2
No
1) Connect a computer to an IONet 3 switch
2) Set the computer IP address to192.168.3. xxx
Re-flash and download to the controller
Are you able to ping 192.168.3 .8 (R)/9 (S)/10 (T) ?
No
Are you able to ping to controllers?
Yes
Yes
No
Finish
Finish
Replace controller



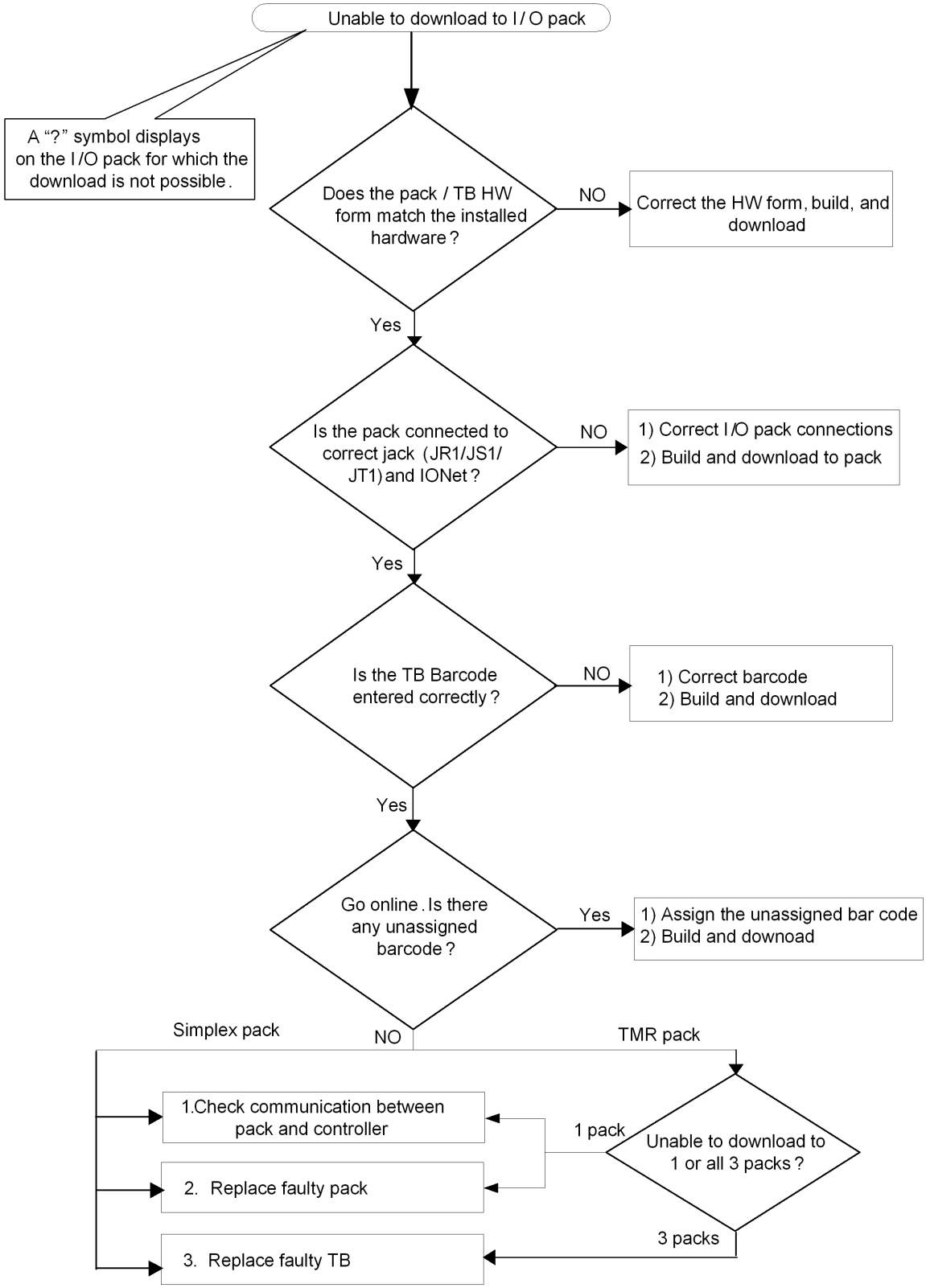
- It is not possible to connect to the designated controller if the controllers are set in the dc determination state. Going online with the individual controllers is required.
- The ToolboxST application may not display any information in the Status tab for the faulty Mark VIe controller if it is set in the Power on/d0-MASTER_INITIALIZATION state. In such a case, the other two controllers are suspended at d1-State Designated Controller Determination state.



Attribute	R Controller	S Controller	T Controller
Control State		Designated Controller Det...	Designated Controller Determination
Controller Equality		Equal	Equal
IO Equality	Equal	Equal	Equal
DDR Equality		Equal	Equal
Designated Controller		Unknown	Unknown
Frame Synchronized		Yes	Yes
UDH Communicator		Yes	Yes
System Idle Time (%)		97.9	98.8
Frame Idle Time (%)		87.1	87.1
Number of Forced Variables		0	0
Heart Beat	0	2151	2521
Controller Time		1/1/1988 1:06:09 PM	1/1/1988 1:06:09 PM
Controller Diagnostic		No	No
I/O Diagnostic		Yes	Yes
Base Load		V02.00.14C	V02.00.14C
Firmware		V03.02.10B	V03.02.10B

- If the controller halts before d0-MASTER_INITIALIZATION (at power-on state), the problem could be with the controller.
- If the controller halts at d0-MASTER_INITIALIZATION state, the controller may have lost its IP address.
- If the controllers halt at the d1-dc_DETERMINATION state, one of the controllers may be unable to communicate to either IONet 1, 2, or 3 port.
- If all the controllers halt at the d3-INPUTS_ENABLED state, at least one of the I/O packs configured as a required module may not be communicating with the controllers.

Unable to Download to I/O Packs



Unable to Communicate with an I/O Pack

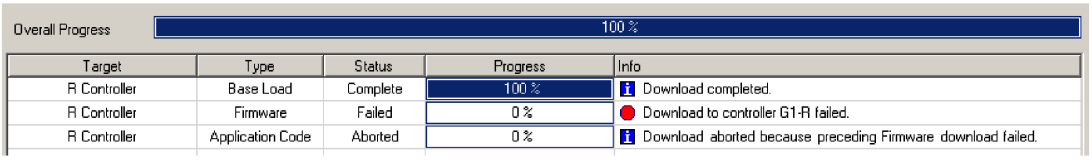
- From the Mark VIe component, Hardware tab, a red X symbol on an I/O pack indicates that the controller is unable to communicate with that I/O pack.
- If multiple I/O packs have red X symbols, the problem may be with the Ethernet switch or the I/O pack power supply.
- Right-clicking an I/O pack with a red X symbol and selecting View Diagnostics displays any I/O communication failure between the I/O pack and the controller diagnostic.

Note In TMR I/O packs, the diagnostic is displayed for the faulty I/O pack only.

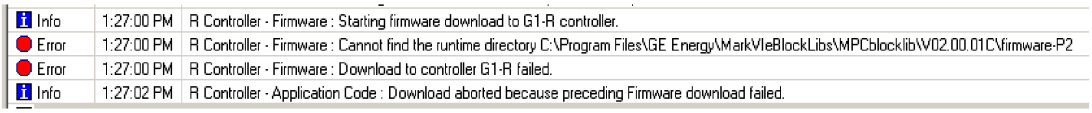
Unable to Download to the Controller or I/O Pack

- If intermittent communication loss occurs between the HMI and the controllers, the downloading process could be interrupted at intermediate stages. It is recommended that the download be carried out for base load, firmware, and application/ parameters individually to the controllers and the I/O packs.
- Firmware download to the controllers could be interrupted due to issues related to incorrect library references.

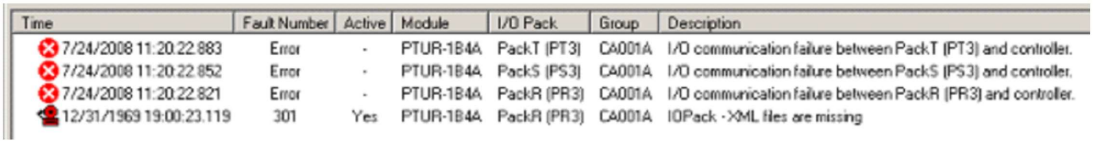
The Eg- MPCBlock libraries are not supported by the UCSA controller. Therefore, if the application is configured for the UCSA controller and the MPCBlockLib is used as a reference library, the firmware download is interrupted.



The following message displays in the Status tab.

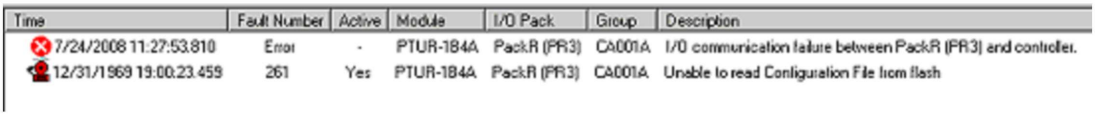


- If there are diagnostic alarms on an I/O pack related to .xml files, the base load, firmware, and parameters must be downloaded to the I/O pack.



Time	Fault Number	Active	Module	I/O Pack	Group	Description
7/24/2008 11:20:22.883	Error	-	PTUR-1B4A	PackT (PT3)	CA001A	I/O communication failure between PackT (PT3) and controller.
7/24/2008 11:20:22.852	Error	-	PTUR-1B4A	PackS (PS3)	CA001A	I/O communication failure between PackS (PS3) and controller.
7/24/2008 11:20:22.821	Error	-	PTUR-1B4A	PackR (PR3)	CA001A	I/O communication failure between PackR (PR3) and controller.
12/31/1969 19:00:23.119	301	Yes	PTUR-1B4A	PackR (PR3)	CA001A	IOPack - XML files are missing

- If an I/O pack is unable to go online with the controller when turning on a Mark VIe control, recycle power to the pack.
- If there are diagnostic alarms on an I/O pack related to configuration files, the base load, firmware and parameters must be downloaded to the pack.

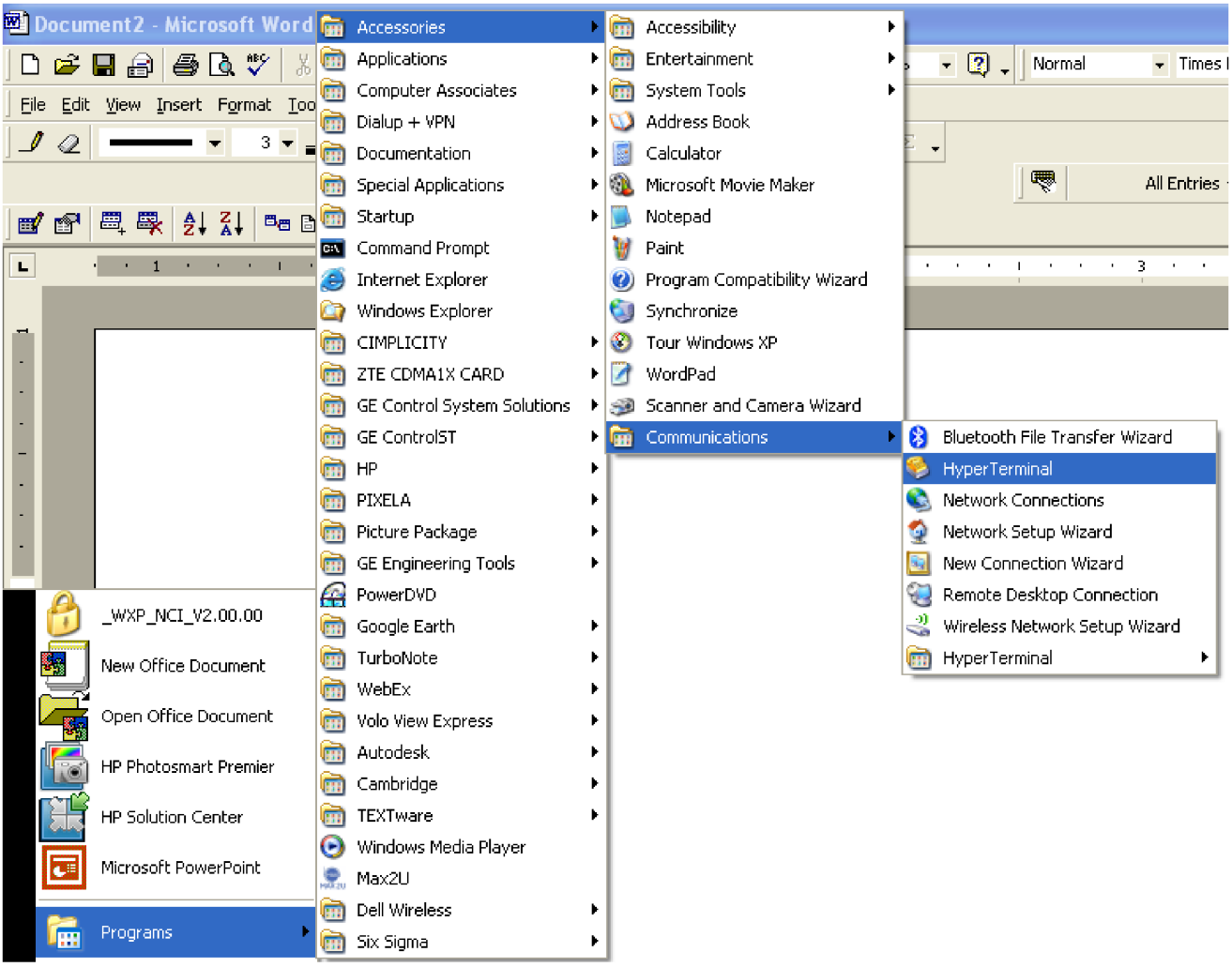


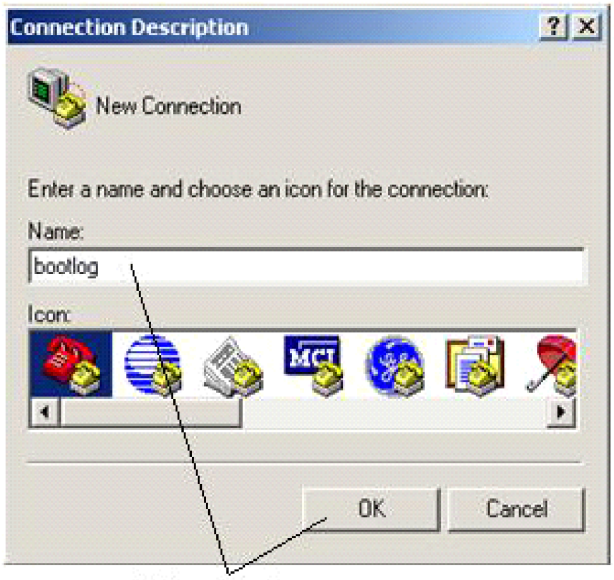
Time	Fault Number	Active	Module	I/O Pack	Group	Description
7/24/2008 11:27:53.810	Error	-	PTUR-1B4A	PackR (PR3)	CA001A	I/O communication failure between PackR (PR3) and controller.
12/31/1969 19:00:23.459	261	Yes	PTUR-1B4A	PackR (PR3)	CA001A	Unable to read Configuration File from flash

Collecting a Boot Log

➢ **To collect a boot log**

1. From the **Start** menu, select **Programs**, **Accessories**, **Communications**, and **HyperTerminal**.



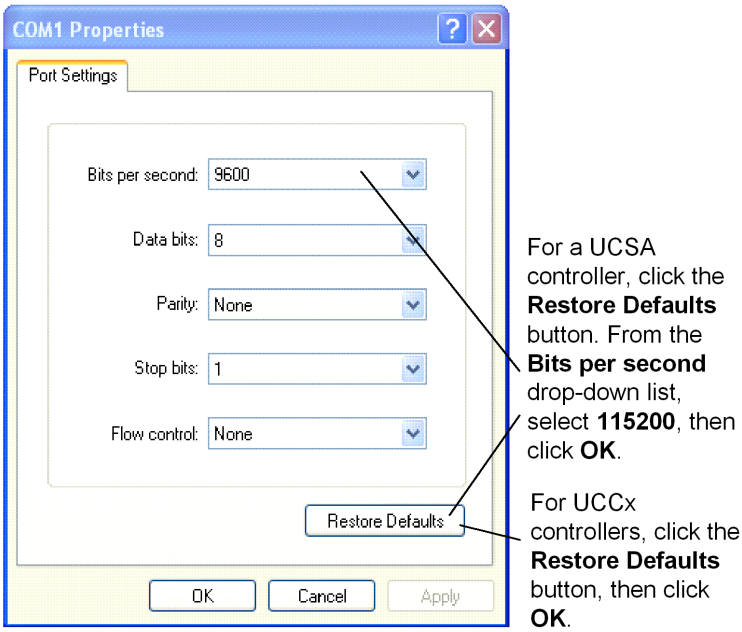


Enter a session name, then click **OK**.



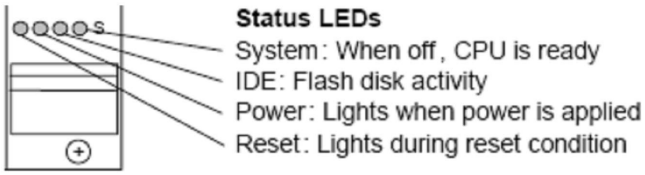
From the **Connect using** drop-down list, select **COM1**, then click **OK**.

Note: The serial cable between the computer COM port and the controller COM port must be connected.

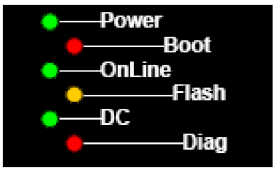


2. Restart the controller, then save the boot log, collected in the HyperTerminal session, with an appropriate file name.

LED Indications



UCCA/ UCCC Status LEDs



UCSA Status LEDs

The UCSA Status LEDs are as follows.

- *Link* displays solid green if the Ethernet PHY on the UCSA has established a link with an Ethernet switch port.
- *Act* indicates packet traffic on an Ethernet interface. This LED may flash if the traffic is low, but is solid green in most systems.
- *Power* displays solid green when the internal 5 V supply is up and regulating. The UCSA converts the incoming 28 V dc to 5 V dc. All other internal power planes are derived from the 5 V.
- *Boot* displays solid red or flashing red during the boot process. The boot flash codes are described below.
- *Online* displays solid green when the controller is online and running application code.
- *Flash* flashes amber when any flash device is being accessed.
- *Dc* displays solid green when the controller is the designated controller.
- *Diag* displays solid red when the controller has a diagnostic available.
- *On* displays solid green when the USB is active.

The UCSA boot LED flash codes are as follows.

The boot LED is on continuously during the boot process unless an error is detected. If an error is detected, the LED flashes at a 1-Hz frequency. The LED, when flashing, is on for 500 ms and off for 500 ms. The number of flashes indicates the failed state. After the flash section, the LED turns off for three seconds.

The blinking codes are as follows.

- 1: Failed Serial Presence Detect (SPD) EEPROM.
- 2: Failed to initialize DRAM or DRAM tests failed.
- 3: Failed NOR flash file system verification.
- 4: Failed to load FPGA or PCI failed.
- 5: CompactFlash device not found.
- 6: Failed to start IDE driver
- 7: CompactFlash image not valid.

If the CompactFlash image is valid but the runtime firmware has not been loaded, the Boot LED flashes continuously at a 1-Hz frequency. When the firmware is loaded, the boot LED turns off.

The I/O pack Status LEDs are as follows.

- A green LED labeled PWR indicates the presence of control power.
- A red LED labeled ATTN indicates I/O pack status, with the following conditions.
  - LED off - no detectable problems with the I/O pack
  - LED solid on - a critical fault is present that prevents the I/O pack from operating. Critical faults include detected hardware failures on the processor or acquisition boards, or no application code is loaded.
  - LED flashing quickly (¼ cycle) - an alarm condition is present in the I/O pack (for example, the wrong I/O pack on the terminal board), or there is no terminal board, or there were errors loading the application code.
  - LED flashing at medium speed (¾ cycle) - the I/O pack is not online.
  - LED flashing slowly (2 cycle) - the I/O pack has received a request to flash the LED indicating that the I/O pack is used during factory test, or as an aid to confirm the physical location against the ToolboxST application settings.
- A green LED labeled LINK is provided for each Ethernet port to indicate that a valid Ethernet connection is present.
- A yellow LED labeled TxRx is provided for each Ethernet port to indicate when the I/O pack is transmitting or receiving data over the port.

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